

TourPhysics: Bringing Physics to World Models for Exploration and Manipulation from a Single Image

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Interactive visual world models must distinguish observation from physical intervention. Camera motion reveals new surfaces, whereas intervention changes object motion, contact, and deformation. Current video world models are largely driven by appearance priors and often lose physical or spatial consistency over long horizons. We present TourPhysics, an online framework initialized from a single image and a declarative physical configuration. TourPhysics extends PhysOmni, our ACM Multimedia 2026 work, from finite physics-grounded video synthesis to persistent exploration and manipulation. TourPhysics combines deterministic simulation with video generation while assigning separate roles to simulator state, geometric evidence, generator controls, and appearance memory. For each action, the simulator computes a finite physical and camera trajectory before the corresponding observation is generated. Accepted observations publish the terminal state and update the appearance memory and subsequent generator controls, while the committed state and simulator geometry remain fixed throughout synthesis and retry. We further separate the simulator geometry used for projection and visibility from the relative depth used to condition the generator. A reference-anchored memory retrieves accepted static appearance through geometric cross-view correspondence and incorporates it through a bounded residual that reverts to the native path when no valid correspondence exists. On simulator-defined camera tours and object manipulations, TourPhysics follows prescribed camera and object trajectories more closely than the evaluated baselines, preserves the input scene, and reduces appearance drift during long-horizon revisits.

Keywords: World models; Physical simulation; Video generation; Single-image scene reconstruction; Persistent memory; Camera control; Physical interaction



1 Introduction

An interactive world model must distinguish observation from intervention. Moving a camera reveals previously unseen surfaces, whereas applying a force changes object positions, contacts, and deformations. Both operations produce new images, but only intervention changes the physical state; observation instead expands the knowledge that the system holds of the same scene.

Recent video generators produce convincing camera motion and object dynamics, and interactive world models increasingly accept viewpoint and action controls [2, 5, 10, 11, 13, 28, 49]. Most of these systems rely on learned appearance priors, so plausible video does not guarantee a persistent or physically faithful world. A scene may change when revisited, temporal caches may carry hallucinated details forward, and learned motion may ignore a prescribed interaction. These failures arise when generated appearance is allowed to stand in for geometric evidence or physical state.

Physics-aware generation methods add trajectories, forces, material controls, or simulator outputs to video synthesis [1, 18, 24]. Geometry-aware methods use camera poses, depth, or reconstructed scene evidence to improve viewpoint control and cross-view consistency [32, 48]. Combining these capabilities requires explicit ownership of state. Our work builds on PhysOmni [54] and extends its single-image, physics-grounded, multi-object scene generation and real-time interaction setting to persistent exploration and manipulation.

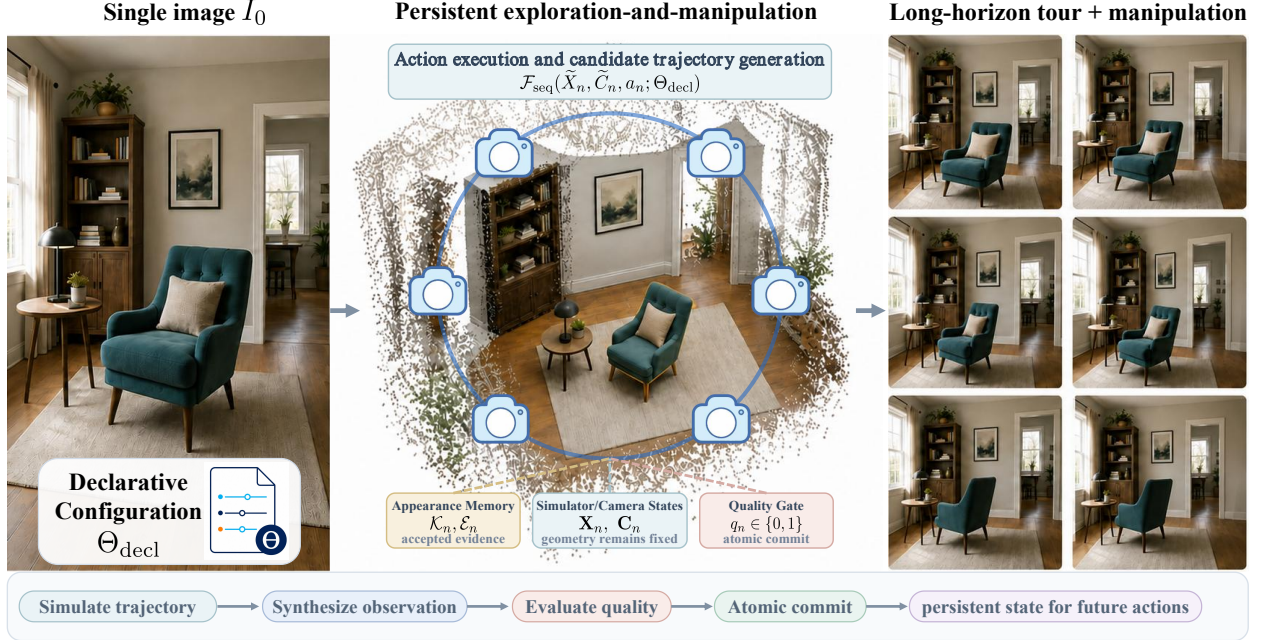


Figure 1 Given a single image I_0 and a declarative configuration Θ_{decl} , TourPhysics maintains a persistent scene model for camera exploration and object manipulation. For each action, a physical transition computes a fixed physical and camera trajectory; the system then synthesizes the corresponding observation, applies a quality gate, and atomically commits the simulator state together with the accepted appearance evidence. The simulator state and geometric record remain fixed during synthesis, and the committed state supports the long-horizon tour and manipulation shown on the right.

While PhysOmni focuses on generating and interacting with a physics-grounded scene from a single image, TourPhysics maintains an explicit world state across multiple action–observation cycles. Each cycle consists of a simulator-defined physical and camera trajectory followed by observation synthesis over that fixed trajectory segment. A quality-gated commit then either publishes the terminal state and the accepted appearance evidence or leaves the persistent world unchanged. In this way, TourPhysics turns physics-grounded generation into an online, stateful world-model loop: the simulator governs object evolution, simulator-consistent geometry governs projection and visibility, and the generator supplies appearance without altering the physical scene model.

Single-image initialization makes this separation especially important. An image constrains visible appearance but leaves occluded geometry, scale, material parameters, and future dynamics underdetermined. TourPhysics combines the image with a user-provided declarative configuration to build one persistent, controllable physical hypothesis. The configuration specifies controllable objects, material families, boundary conditions, interventions, and camera actions.

TourPhysics processes one committed observation window at a time. For each action, it computes a finite physical and camera trajectory and then generates an observation of that fixed segment. A quality gate either commits the terminal state and the appearance evidence together or leaves the persistent state unchanged. Generated observations may extend appearance knowledge, but they cannot modify the committed dynamics or the simulator geometry.

Figure 1 summarizes this workflow, from a single image to an interactive world, and illustrates the resulting long-horizon camera exploration and object manipulation.

TourPhysics assigns different roles to two depth representations. Simulator-consistent depth governs projection, visibility, correspondence, and trajectory evaluation, while relative depth conditions the video generator and may be completed from accepted observations. For long-horizon consistency, the system retains the original input image as a permanent reference and retrieves the accepted appearance through geometric cross-view correspondence. Historical evidence enters attention as a bounded residual; when no valid correspondence is available, the model follows its native reference-conditioned path.

We evaluate TourPhysics on simulator-defined camera tours and object manipulations involving rigid, cloth, elastic, granular, and fluid proxies. The evaluation measures alignment with prescribed camera and object trajectories, event timing and deformation, input-scene similarity, and long-horizon appearance stability.

Our contributions are twofold:

- We formulate single-image tour-and-manipulate generation as coupled physical and epistemic transitions and present TourPhysics as an online world model with explicit state ownership. The simulator fixes the physical and camera trajectory before observation synthesis, while a quality-gated commit publishes the accepted state and visual evidence atomically.
- We introduce two complementary consistency mechanisms: simulator-consistent geometry is kept separate from generator-facing depth, and a reference-anchored memory retrieves accepted static appearance through geometry-routed cross-view correspondence. Historical features enter as a confidence-bounded residual and fall back exactly to the native reference-conditioned path when correspondence is invalid.

2 Related Work

2.1 Video World Models and Interactive Environments

Diffusion models supply strong appearance and motion priors for video generation [2, 4, 5, 28, 29, 58, 60, 91], while recent analyses of foundation models summarize the scaling and conditioning trends underlying these systems [59]. Complementary AI Flow studies consider system-level AI service organization [81, 82]. Studies of temporal prediction and interpolation further reveal the need to preserve motion and appearance across adjacent frames [61, 62], whereas interactive world models synthesize observations conditioned on learned actions or compact state representations [10, 12–15, 83]. Playable Environments represents a scene from a single image through controllable camera poses and learned object actions [16], whereas Dexterous World Models conditions video diffusion on a static 3D rendering together with an egocentric hand trajectory [17]. Although these methods support action-controlled video synthesis, they generally do not distinguish observation from intervention within their state updates. In TourPhysics, only the physical branch updates the simulator state, whereas camera-induced observations update appearance knowledge only after they commit.

2.2 Geometry-Aware Video and Persistent Scene Representations

Camera-controlled video generation relies on camera pose, depth, optical flow, or reconstructed 3D evidence to improve viewpoint control and cross-view consistency [32, 35, 36, 70, 74, 75, 93]. Persistent approaches maintain point clouds or explicit 3D/4D representations across generated views [33, 34, 37, 38, 66, 69, 71]. Single-image methods estimate camera parameters and dense geometry, or construct object-centric and scene-level representations [31, 41–47, 67, 68, 72, 73, 84, 85, 92]. CUT3R is closely related, as it maintains a shared metric pointmap as observations arrive [39]. Nevertheless, monocular scale, topology, and occlusion remain ambiguous [40]. TourPhysics uses visible geometry together with a declarative configuration to initialize a controllable physical hypothesis, reserves simulator depth for projection and visibility, and employs separate depth to condition the generator.

2.3 Physics-Grounded and Simulator-Guided Generation

Physics-aware generation incorporates learned dynamics, trajectories, forces, material controls, or explicit simulation into image and video synthesis [1, 18, 20–25, 90]. Related TPAMI work on physics-based generative restoration couples image priors with explicit degradation constraints [76], while masked operators such as partial convolution restrict updates to the valid support during inpainting and synthesis [77]. PhysGen3D constructs an amodal, camera-centric world from user-specified initial conditions and material properties [19]. Physical Simulator In-the-Loop Video Generation reconstructs a generated template, simulates foreground trajectories, and uses simulator correspondences to guide synthesis [26]. TourPhysics instead computes each

declared action with a rigid-body, PBD [3], or MPM [27] model before the video model observes the resulting fixed state sequence.

2.4 Long-Video Memory and Reference Conditioning

Long-video methods extend temporal support through sliding windows, recurrent contexts, or bounded attention caches. Related autoregressive work improves long-horizon planning, test-time correction, and attention efficiency [86–89]. Reference-conditioned and memory-based approaches transfer texture or recover long-range context through correspondence mechanisms, dynamic memory, and cross-attention [8, 9, 63]. Established surveys on visual tracking provide complementary correspondence and temporal-association baselines for maintaining identity under motion and occlusion [64, 65]. LongVie supports multimodal long-video generation [6], whereas Deep Forcing introduces a training-free cache rule to retain early context under finite attention capacity [7]. Although these mechanisms improve temporal continuity, they do not explicitly associate historical appearance with persistent geometric surfaces or committed simulator states. TourPhysics augments a finite-window video diffusion model with geometry-routed memory, in which an accepted historical key/value page is selected through cross-view visibility and incorporated as a confidence-bounded residual around the permanent real-image reference.

3 Problem Formulation

Given a single RGB image I_0 and a declarative configuration Θ_{decl} , we aim to construct an online world model that supports both camera exploration and physical manipulation. The desired output is a long-horizon sequence of observations that follows the declared physical–camera trajectory while maintaining appearance consistency through committed state transitions across accepted windows. The formulation is organized into three parts: Figure 2(1), *Scene Lifting (One-time Preprocessing)*, which constructs the persistent scene hypothesis and reference; Figure 2(2), *Coupled Physical and Epistemic Transitions*, which advances a single committed window; and Figure 2(3), *Long-Horizon Tour-and-Manipulate Video (Output)*, which produces the resulting observation sequence.

This separation makes the ownership of state explicit: the physical branch determines the simulator state and geometry, the observation branch synthesizes observations for the resulting fixed segment, and the epistemic branch publishes only accepted appearance evidence. The resulting causal order is illustrated in the bottom protocol of Figure 1: simulate the trajectory, synthesize the corresponding observation, evaluate the quality gate, and atomically commit the accepted result for subsequent actions. In particular, the committed simulator and camera states serve as the starting point for the next action, while the accepted appearance memory and tail evidence provide conditioning context for subsequent generation.

3.1 Scene Lifting and Persistent Reference

The input is given by

$$\mathcal{I}_{\text{init}} = (I_0, \Theta_{\text{decl}}),$$

where Θ_{decl} specifies the controllable objects, material and physical-model assignments, boundary conditions, interventions, and camera trajectory controls. The one-time scene-lifting stage constructs an internally consistent scene hypothesis from this underdetermined input, estimates a reference camera and calibrated geometry, and preserves I_0 , together with any cached background plate, as permanent reference information. These reference records and the declaration remain fixed throughout online execution and are read by every subsequent operator.

At the boundary preceding the execution of action $a_n = (c_n, u_n)$, the committed state is given by

$$\mathcal{S}_n = (X_n, C_n, \mathcal{K}_n, \mathcal{E}_n).$$

The index n counts the accepted windows. A rejected attempt is treated as a retry of the same action, indexed by r , and leaves both n and $\mathcal{S}_n$ unchanged; for notational simplicity, we suppress r below.

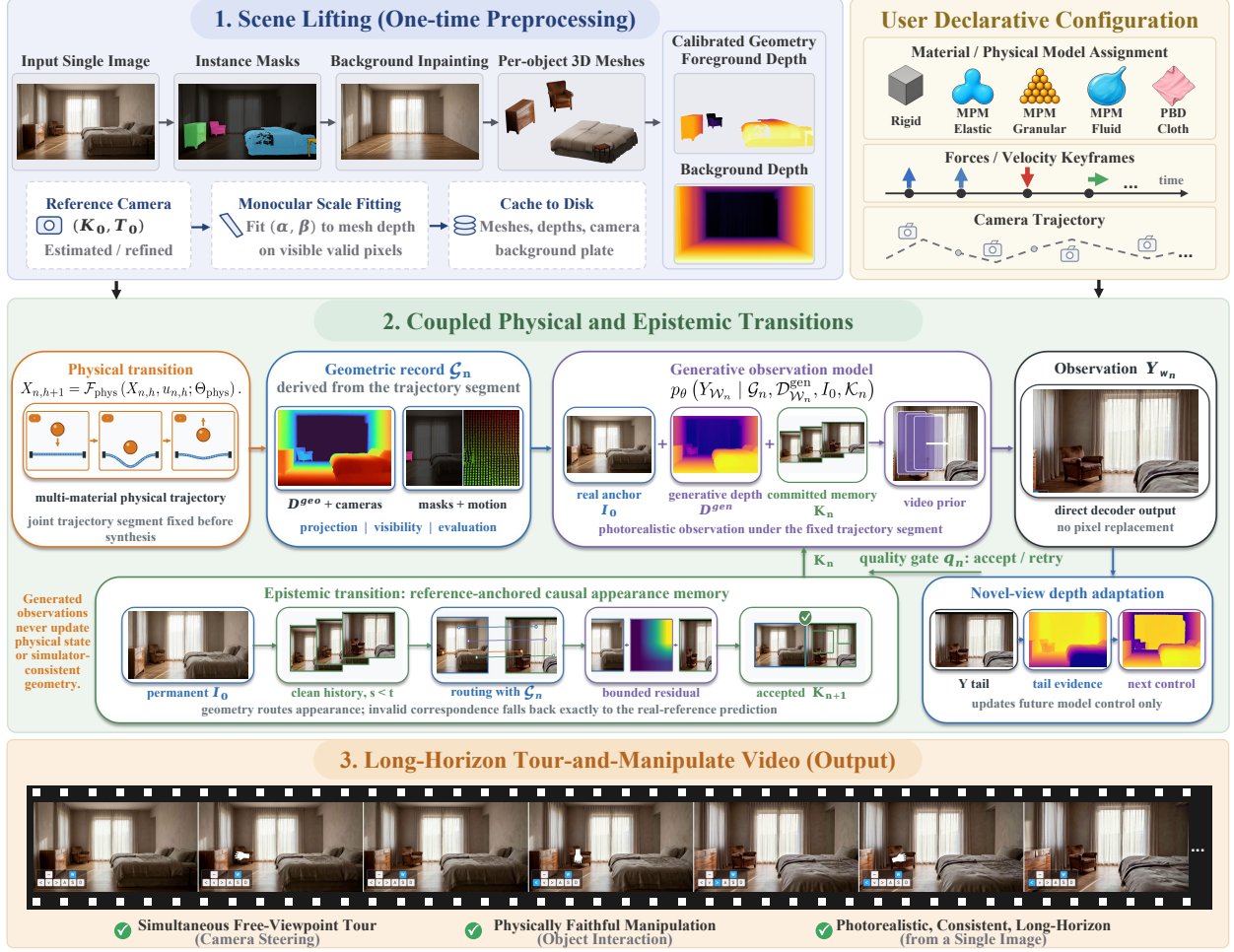


Figure 2 Overview of TourPhysics. (1) *Scene Lifting (One-time Preprocessing)* combines the input image with a declarative user configuration, including material models, force and velocity keyframes, and camera commands, to construct a persistent scene hypothesis comprising instance masks, a background plate, per-object 3D meshes, calibrated foreground and background geometry, a reference camera, and cached simulator assets. (2) *Coupled Physical and Epistemic Transitions* advances one action window. The orange *Physical transition* block fixes the multi-material physical and camera segment before synthesis; the read-only geometric record $\mathcal{G}_n$ supplies simulator-consistent depth, cameras, masks, and motion; and the observation model combines these controls with generator-facing depth $\mathcal{D}_{W_n}^{\text{gen}}$, the real-image anchor I_0 , and committed appearance memory $\mathcal{K}_n$ to produce Y_{W_n} . A quality gate publishes the accepted state and appearance atomically; only after acceptance do reference-anchored memory and novel-view depth adaptation affect subsequent appearance controls. (3) *Long-Horizon Tour-and-Manipulate Video (Output)* chains the accepted windows into a persistent sequence that supports free-viewpoint exploration, physically faithful manipulation, and photorealistic long-horizon consistency, while the generated RGB never replaces simulator renderings or writes back to the physical state or simulator geometry.

3.2 Coupled Physical and Epistemic Transitions

Physical transition and geometric record. At a committed boundary, the camera command c_n and the optional intervention u_n are resolved into a framewise physical-camera trajectory before any pixels are sampled. The simulator and camera are advanced from working copies of the preceding committed checkpoint,

$$(\tilde{X}_n, \tilde{C}_n) = \text{clone}(X_n, C_n),$$

so that a candidate attempt produces no persistent side effects.

The physical and camera trajectories, together with the associated geometric record, are then produced

by

$$(\mathbf{X}_n, \mathbf{C}_n, \mathcal{G}_n) = \mathcal{F}_{\text{seq}} \left(\tilde{X}_n, \tilde{C}_n, a_n; \Theta_{\text{decl}} \right). \quad (1)$$

Here,

$$\begin{aligned} \mathbf{X}_n &= \left(X_n^{(0)}, X_n^{(1)}, \dots, X_n^{(T_n)} \right), \\ \mathbf{C}_n &= \left(C_n^{(0)}, C_n^{(1)}, \dots, C_n^{(T_n)} \right) \end{aligned}$$

denote the framewise physical and camera state sequences, respectively. The index $t = 0$ denotes the committed boundary state, while $t = 1, \dots, T_n$ index the newly generated states within the candidate window:

$$X_n^{(0)} = \tilde{X}_n, \quad C_n^{(0)} = \tilde{C}_n.$$

The terminal working states are given by

$$X_n^+ = X_n^{(T_n)}, \quad C_n^+ = C_n^{(T_n)}.$$

The candidate record $\mathcal{G}_n$ contains simulator-consistent framewise depth, camera parameters, object masks, motion, visibility, correspondence, and validity information derived from $(\mathbf{X}_n, \mathbf{C}_n)$.

Generative observation model. Once the candidate window $\mathcal{W}_n$ and its geometric record have been fixed, the observation model takes as input the geometric record, the generator-facing depth $\mathcal{D}_{\mathcal{W}_n}^{\text{gen}}$, the permanent real-image anchor I_0 , and the committed appearance memory $\mathcal{K}_n$, and produces a candidate observation.

Prior to sampling, the generator-facing depth is constructed from the fixed geometric record and the accepted tail evidence:

$$\mathcal{D}_{\mathcal{W}_n}^{\text{gen}} = \mathcal{F}_{\text{ctrl}}(\mathcal{G}_n, \mathcal{E}_n).$$

Once constructed, the full framewise control tensor remains fixed for the duration of the current candidate window and is not recomputed during observation sampling:

$$Y_{\mathcal{W}_n} \sim p_{\theta}(Y_{\mathcal{W}_n} \mid \mathcal{G}_n, \mathcal{D}_{\mathcal{W}_n}^{\text{gen}}, I_0, \mathcal{K}_n). \quad (2)$$

Because $\mathcal{G}_n$ encodes the simulator-consistent framewise geometric and camera records derived from the physical and camera trajectories, the trajectory information required by the observation model is supplied implicitly through $\mathcal{G}_n$. Although $\mathcal{D}_{\mathcal{W}_n}^{\text{gen}}$ is deterministically derived from $\mathcal{G}_n$ and $\mathcal{E}_n$, it is supplied explicitly because it constitutes the representation consumed by the generator. The conditional form above therefore specifies the generative observation model.

The real-image anchor I_0 is immutable and is used solely as a conditioning reference. Regions that are not observed in I_0 , including regions exposed by the current camera trajectory, may be synthesized using the fixed geometric record, the accepted tail evidence, and the committed appearance memory. Simulator-derived signals may provide conditioning, geometric, and evaluation information, but simulator RGB is not used as a direct substitute for generated pixels. Consequently, the observation branch may extend the appearance knowledge through accepted generated observations, but it cannot modify X_n , C_n , or the simulator-consistent geometry that produced the current candidate window.

After sampling, the candidate observation is used to construct the staged appearance memory and staged tail evidence:

$$(\hat{\mathcal{K}}_{n+1}, \hat{\mathcal{E}}_{n+1}) = \mathcal{U}_{\text{stage}}(\mathcal{K}_n, \mathcal{E}_n, Y_{\mathcal{W}_n}, \mathcal{G}_n). \quad (3)$$

These staged quantities are candidate-local and remain private until the candidate window passes the quality gate. If the candidate is rejected, they are discarded together with the working trajectory, and the committed appearance memory and accepted tail evidence remain unchanged.

Epistemic transition and quality-gated commit. The candidate appearance memory and tail evidence remain private while the current window is being evaluated. A binary quality decision is computed as

$$q_n = Q(Y_{\mathcal{W}_n}, \mathcal{G}_n, \mathcal{D}_{\mathcal{W}_n}^{\text{gen}}, \mathcal{K}_n, \mathcal{E}_n) \in \{0, 1\}, \quad (4)$$

which controls the atomic state transition. Here, $\hat{\mathcal{K}}_{n+1}$ and $\hat{\mathcal{E}}_{n+1}$ denote the staged memory and tail-evidence updates proposed by the current candidate observation:

$$\begin{aligned} q_n = 1 : \quad \mathcal{S}_{n+1} &= (X_n^+, C_n^+, \hat{\mathcal{K}}_{n+1}, \hat{\mathcal{E}}_{n+1}), \\ q_n = 0 : \quad \mathcal{S}_n &\text{ remains unchanged.} \end{aligned} \quad (5)$$

An accepted window atomically publishes the terminal simulator and camera states together with the staged visual evidence. A rejected attempt discards its working copy and leaves the committed checkpoint $\mathcal{S}_n$ unchanged. A retry retains the same action, trajectory, geometric record, generator-facing controls, and committed memory, while allowing only a new sampling seed. The epistemic branch therefore updates $\mathcal{K}_{n+1}$ and the evidence used for future controls only after acceptance.

3.3 Long-Horizon Tour-and-Manipulate Video (Output)

The final output is the sequence of committed observations. To avoid duplicated boundary frames, the first frame of every subsequent window is removed prior to concatenation:

$$\text{Video} = \text{concat}(Y_{\mathcal{W}_0}, Y_{\mathcal{W}_1}[1:], \dots, Y_{\mathcal{W}_{N-1}}[1:]). \quad (6)$$

Here, N denotes the total number of accepted windows; rejected attempts do not contribute output frames.

The resulting video must support both free-viewpoint camera exploration and physically faithful object manipulation while remaining photorealistic and consistent over long horizons. These requirements are evaluated with respect to the declared trajectory and the persistent committed state, rather than through unconstrained visual continuation. The implementation of each operator is described in Section 4.

4 Methods

4.1 Overview

The method follows the causal order of *scene lifting*, *physical/camera transition*, *geometric record*, *observation synthesis*, *staged update*, *quality gate*, and *atomic commit* (Fig. 2). Scene lifting builds one editable hypothesis from I_0 and Θ_{decl} ; at each accepted boundary, the simulator and camera controller clone the committed state and produce the complete trajectory and the read-only record $\mathcal{G}_n$. The physical segment and its geometry are fixed before RGB sampling, and a retry reuses them.

The observation model then consumes $\mathcal{G}_n$, a separate generator-facing depth control, the permanent image anchor, and the committed appearance context. Memory and tail-depth updates remain candidate-local until the quality gate; a passing window commits the terminal simulator/camera state and the staged evidence together, while a failing window rolls back the candidate. Generated RGB can therefore extend appearance knowledge without modifying physical state or simulator geometry.

4.2 Scene Lifting and Persistent Reference

Scene lifting is performed once, prior to the first interaction. Given the input image I_0 and the frozen declaration Θ_{decl} , the initializer segments the selected objects, lifts their masks to explicit meshes, removes the foreground to form a cached background plate B_0 , and aligns the meshes to a shared gravity-aware scene frame. A reference camera and an affine depth calibration are then fitted from the visible image evidence. These operations define a single controllable physical hypothesis; they are not repeated as online state corrections.

Object decomposition and background. For each object selected by the user, SAM 3 and SAM 3D Objects produce an instance mask, a local triangular mesh, and an object-to-camera similarity transform. These explicit surfaces, rather than the RGB image, serve as the contact and collision interfaces used by the simulator. The selected foreground union is removed cumulatively with LaMa inpainting [62, 77] to form the cached plate B_0 , while the unmodified I_0 is retained as the permanent appearance anchor. The plate is used only when a camera move exposes background support; it is not treated as a new physical object.

The independently lifted meshes are placed in a shared gravity-aware frame. We pool the lowest 5% of vertices as provisional contact anchors (relaxed to 10% when an object has too few anchors), fit a robust ground plane, and apply one bounded rigid alignment. Small residual overlaps are corrected along the minimum-overlap bounding-box axis. These corrections belong to the initial hypothesis and are never applied online.

The declaration may be authored directly or proposed by the optional vision-language initializer. In either case, the material and force-field entries are checked against the supported families, completed with defaults, exposed for editing, and frozen before the first trajectory simulation. We use the fixed projections $\Theta_{\text{phys}} = \Pi_{\text{phys}}(\Theta_{\text{decl}})$ and $\Theta_{\text{cam}} = \Pi_{\text{cam}}(\Theta_{\text{decl}})$. An online action selects a camera command and an intervention for the next finite window; it does not replace the declaration.

The reference camera is refined by aligning the rendered union silhouette with the selected masks. For valid pixels, the monocular depth is placed into the simulator metric convention with a single affine fit,

$$D_0^{\text{geo}}(p) = \alpha \tilde{D}_0(p) + \beta, \quad (7)$$

where invalid and non-finite samples are excluded. Depth/normal estimation and sparse-depth completion studies motivate treating this fit as a calibrated geometric control rather than a unique reconstruction [67, 68]; single-image novel-view work likewise highlights the disocclusion ambiguity that remains after calibration [70]. This calibration fixes the scale used by later projection and visibility tests.

The immutable reference record is

$$\mathcal{R} = (I_0, B_0, D_0^{\text{geo}}, V_0, Q_0, L_0, K_0, T_0). \quad (8)$$

Together with the initialized physical and camera states, it defines $\mathcal{S}_0 = (X_0, C_0, \mathcal{K}_0, \mathcal{E}_0)$, where $\mathcal{K}_0$ contains no generated pages and $\mathcal{E}_0$ contains no accepted tail evidence. The reference image, its calibrated geometry, and the cached background plate remain read-only throughout online execution. Camera intrinsics K_t and the appearance memory $\mathcal{K}_n$ are distinct quantities.

4.3 Physical and Camera Transition

At the boundary of accepted window n , the committed state is $\mathcal{S}_n = (X_n, C_n, \mathcal{K}_n, \mathcal{E}_n)$. For action $a_n = (c_n, u_n)$, the simulator and camera controller first clone the committed physical and camera states. The fixed candidate trajectory and its geometric record are then produced by the sequence operator already defined in Eq. (1): $(\mathbf{X}_n, \mathbf{C}_n, \mathcal{G}_n) = \mathcal{F}_{\text{seq}}(\tilde{X}_n, \tilde{C}_n, a_n; \Theta_{\text{decl}})$. The terminal candidates X_n^+ and C_n^+ are the endpoints of this working segment.

Within this fixed segment, each simulator substep applies the declared physical map to the working state, while the camera controller advances in parallel from c_n . The first substep starts at the cloned committed boundary, and the terminal candidate is taken only at the declared sampled endpoint. Keeping this boundary convention explicit ensures that a retry reproduces the same physical trajectory instead of inadvertently advancing it twice.

Camera coverage along the fixed update is illustrated in Fig. 4; the point cloud is diagnostic and does not modify simulator geometry or physical state.

The declaration routes rigid and articulated proxies to RBD, continuum and volumetric materials to MPM, and thin or constraint-dominated systems to PBD. Figure 3 summarizes these material families and their solver routing. Gravity, scheduled force fields, and ongoing dynamics continue even when $u_n = \emptyset$, so a camera-only action can still advance the physical state. RBD-MPM and RBD-PBD contacts are coupled within the same scene step. Unless overridden by the declaration, the solver step is 4 ms, with 10 substeps for rigid-only scenes and 16 for particle scenes; gravity is $(0, 0, -9.8)$. Constitutive equations and material-specific parameters are reported in the supplementary appendix, Sec. A.2 (“Declarative Configuration and Physical Simulation”), as they do not affect the observation transaction.



Figure 3 Material families and solver routing used by the physical backend. The declarative configuration selects the material model and its parameters; the image is a schematic taxonomy, not a generated-video result.

At each frame, the renderer exports the simulator-consistent metric depth, valid-visible support $V_{n,t}$, confidence $Q_{n,t}$, object labels $L_{n,t}$, the foreground mask $M_{n,t}^{\text{mask}}$, motion $U_{n,t}^{\text{mot}}$, camera intrinsics, and world-to-camera extrinsics. These fields form $\mathcal{G}_n$ and serve as read-only conditioning and evaluation signals; they never replace decoded RGB. The reported profile uses 81 decoded frames with one shared conditioning boundary and 80 new frames. The exact simulator-to-video index convention is given in the supplementary appendix, Sec. A.4 (“Observation Generation and Transactional State”); the endpoint convention introduced there also defines X_n^+ and C_n^+ . The complete segment is held fixed during both sampling and retry.

For clarity, the record used by the generator and by the later memory tests is

$$\mathcal{G}_n = \left\{ \begin{array}{l} D_{n,t}^{\text{geo}}, V_{n,t}, Q_{n,t}, L_{n,t}, \\ M_{n,t}^{\text{mask}}, U_{n,t}^{\text{mot}}, K_{n,t}, T_{n,t} \end{array} \right\}_{t \in \mathcal{W}_n}. \quad (9)$$

It is rendered once from the fixed trajectory and is never updated from generated frames. With $f_{\text{sim}} = 60$ and $f_{\text{gen}} = 16$, the center-of-bin mapping used in the evaluated profile is

$$k_n(j) = o_n + \left\lfloor \frac{(2j+1)f_{\text{sim}}}{2f_{\text{gen}}} \right\rfloor, \quad o_{n+1} = o_n + 300. \quad (10)$$

Thus $k_n(80) = k_{n+1}(0)$ is the shared boundary; X_n^+ and C_n^+ are defined at this sampled endpoint rather than at the raw origin $o_n + 300$. This convention removes the off-by-one ambiguity between simulator substeps and decoded frames.

4.4 Geometric Record and Generator Controls

The method keeps the simulator geometry separate from the depth representation consumed by the video generator. $D_{n,t}^{\text{geo}}$ is the metric camera-space z -depth in the calibrated scene units and is written only by scene initialization and simulator rendering. It controls projection, visibility, correspondence, and trajectory evaluation. In contrast, $\mathcal{D}_{\mathcal{W}_n}^{\text{base}} = \mathcal{F}_{\text{base}}(\mathcal{G}_n; \mathcal{R})$ is a normalized relative-depth condition for the video model. This explicit split follows the broader practice of keeping metric reconstruction and appearance-oriented representation separate in point-cloud, dense-reconstruction, and radiance-field systems [66, 69, 74, 75].

For $t \in \mathcal{W}_n$, we retain the three quantities separately:

$$D_{n,t}^{\text{geo}}, \quad D_{n,t}^{\text{base}} = [\mathcal{D}_{\mathcal{W}_n}^{\text{base}}]_t, \quad D_{n,t}^{\text{gen}} = [\mathcal{D}_{\mathcal{W}_n}^{\text{gen}}]_t. \quad (11)$$

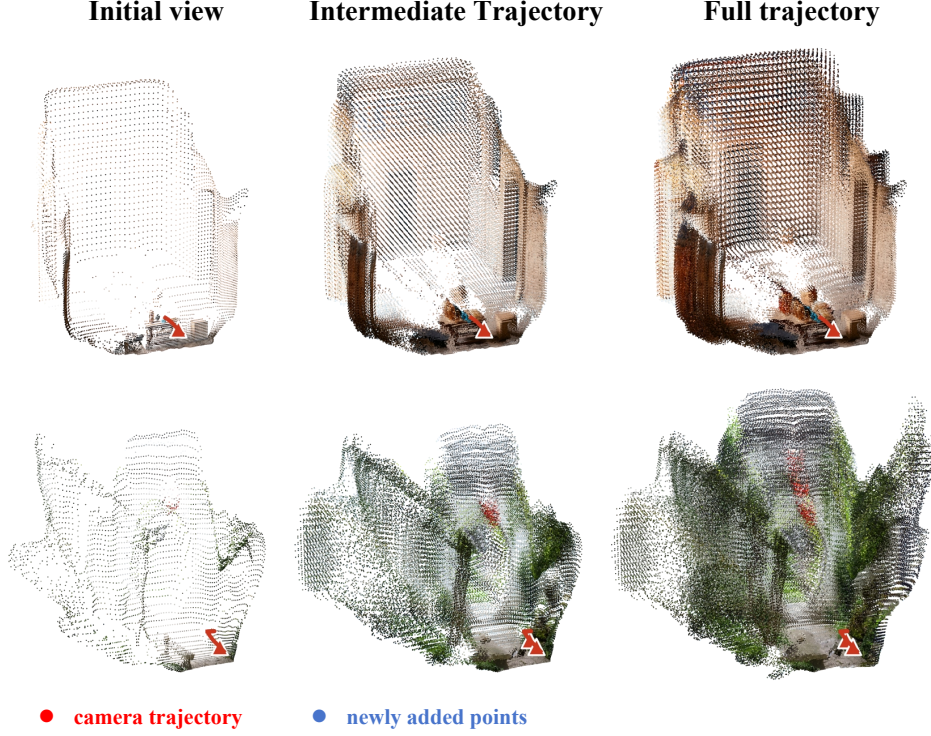


Figure 4 Point-cloud visualization of the camera update. Columns show the initial, intermediate, and full-trajectory views; red arrows denote camera motion and blue points denote newly observed support.

The first is the metric camera-space z -depth in calibrated scene units. The other two are normalized controls with no physical units. Only the first may drive projection, visibility, correspondence, or trajectory evaluation; only the last is supplied to the video generator.

Accepted tail evidence $\mathcal{E}_n$ can complete only the newly exposed, under-supported regions of the current generator control. The transferred evidence is aligned to the fixed control and applied through

$$\mathcal{D}_{\mathcal{W}_n}^{\text{gen}} = (1 - W_n) \odot \mathcal{D}_{\mathcal{W}_n}^{\text{base}} + W_n \odot \overline{\mathcal{D}}_{\mathcal{W}_n}(\mathcal{E}_n), \quad (12)$$

where W_n is zero outside the pointwise eligibility tests. These tests use only $\mathcal{G}_n$ and the evidence accepted before the current sampling call; a missing overlap, a degenerate affine fit, or a failed agreement test sets the affected weights to zero and preserves the base control exactly. Exact alignment rules and thresholds are given in the supplementary appendix, Sec. A.3 (“Separated Depth Controls”). This branch changes visual conditioning only: it never writes D^{geo} , collision geometry, or physical state.

Figure 5 visualizes the four terms and the point-wise routing: the fixed base control, aligned accepted tail evidence, the eligibility mask, and the fused generator control.

The eligibility test is deliberately fail-closed. A candidate tail frame must achieve at least 0.20 trusted static overlap, a Spearman agreement of at least 0.70, a model-space NRMSE of at most 0.08, and a metric-space NRMSE of at most 0.12. Completion is considered only when the median newly visible fraction is at least 0.05, the target hole fraction is at least 0.01, and the trusted source coverage is at least 0.20; at least two of the configured tail frames must pass. Affine alignment uses the overlap samples and is discarded when the fit is degenerate.

The mask W_n is constructed after these frame-level tests and is zero on metric-supported or dynamic pixels. A temporal union followed by a small dilation yields one stable support mask for the entire 81-frame window, rather than a separately moving hole boundary in each frame. Accepted evidence stores only aligned generator-depth estimates, support indices, and confidence statistics. It is never interpreted as a new measurement of simulator depth.

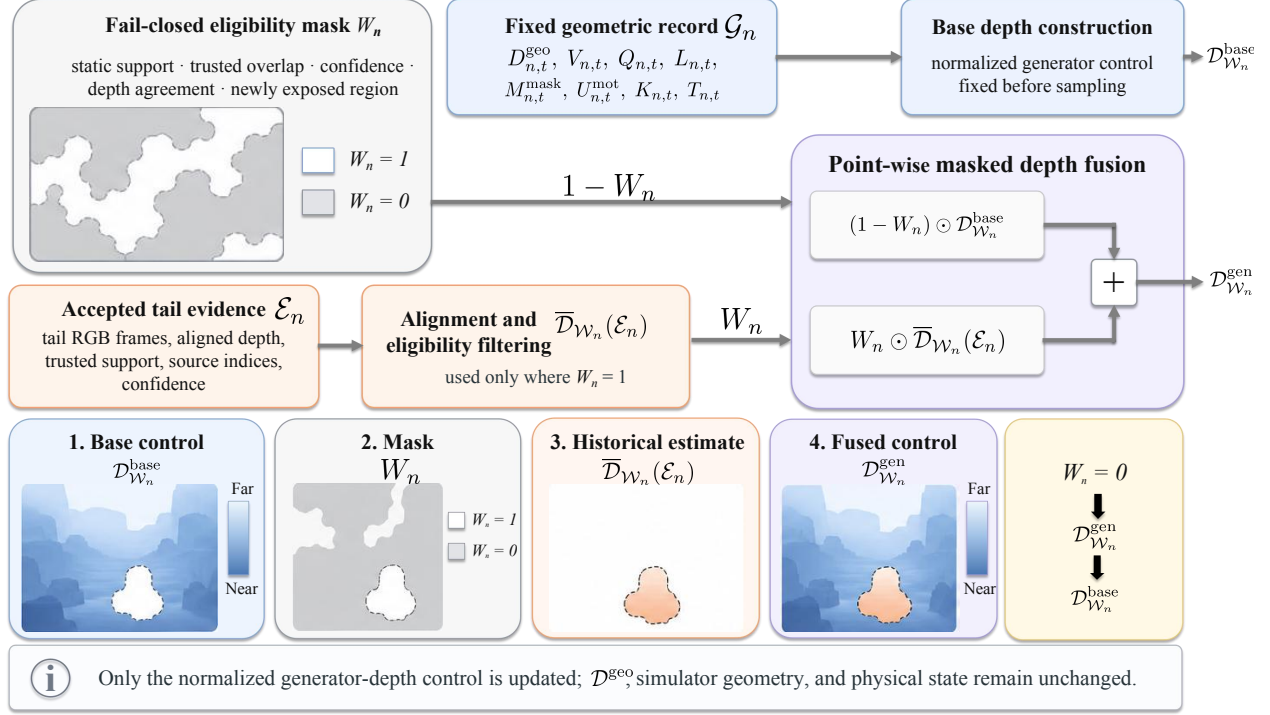


Figure 5 Generator-facing depth update, following the reference-style routing diagram. The fixed geometric record constructs the base control; accepted tail evidence is aligned and filtered into $\overline{\mathcal{D}}(\mathcal{E}_n)$, while W_n gates the newly exposed, under-supported region. The final control is formed by point-wise fusion; outside the mask, the base control is preserved exactly. The depth maps are illustrative visualizations of the update, not additional simulator geometry.

4.5 Observation Synthesis and Staged Update

Once $\mathcal{G}_n$ and $\mathcal{D}_{W_n}^{\text{gen}}$ are fixed, the finite-window diffusion model synthesizes the observation

$$Y_{W_n} \sim p_{\theta}(Y_{W_n} | \mathcal{G}_n, \mathcal{D}_{W_n}^{\text{gen}}, I_0, \mathcal{K}_n). \quad (13)$$

The physical trajectory, camera poses, geometric record, and generator controls remain fixed throughout this call. The permanent image anchor fixes appearance, the current trajectory determines layout, and the committed memory supplies only accepted historical appearance.

Controls are prepared at 640×352 , while the decoded output is 832×480 at 16 fps. An 81-frame window contains one shared conditioning boundary and 80 new frames; consecutive windows therefore advance by 80 decoded frames. The finite native cache handles adjacent temporal continuity, whereas the geometry-routed pages described below provide sparse long-range access, complementing the temporal prediction and interpolation mechanisms used in prior video models [60, 61]. The abstract state records the long-range pages as $\mathcal{K}_n$; the native cache is checkpointed together with the same observation context. Model weights, the denoising schedule, and the camera/physics controls remain frozen during this call.

The conditioning record is assembled before sampling and is not recomputed from intermediate denoising states. In particular, simulator renderings serve as control and evaluation signals rather than replacement RGB. This distinction prevents a successful-looking simulator frame from silently becoming the generated observation and keeps the observation branch from acquiring physical authority.

The memory branch retains reliable static support from earlier committed views. Geometry-routed correspondence uses $\mathcal{D}^{\text{geo}}$, camera parameters, visibility, confidence, and static-support masks to select at most one eligible source page for each target location. Moving or deforming proxies are excluded. The design is related to memory-based video segmentation and tracking systems that use explicit temporal correspondence, but here the eligibility test is additionally tied to persistent scene geometry [63, 64]. Only

visible, high-confidence static support enters the long-range memory:

$$\Omega_t = V_t \cap M_t^{\text{sta}} \cap \{Q_t \geq \tau_q\} \cap \neg \Gamma_t^{\text{edge}}, \quad (14)$$

where M_t^{sta} is derived from the declaration and the motion/deformation field, and Γ_t^{edge} removes unreliable depth boundaries. For a target pixel p , the metric depth back-projects it into the shared scene frame, and an earlier source view receives the corresponding projection. Here T_t denotes the world-to-camera transform, and π^{-1} returns a camera-frame point with the supplied metric z -depth:

$$x_w = T_t^{-1} \pi^{-1}(p, D_t^{\text{geo}}(p); K_t), \quad (\hat{p}_s, \hat{z}_s) = \pi(T_s x_w; K_s). \quad (15)$$

The source must be in bounds, visible, static, and depth-consistent. Its confidence is

$$c_s(p) = Q_t(p) Q_s(\hat{p}_s) \exp\left(-\frac{|D_s^{\text{geo}}(\hat{p}_s) - \hat{z}_s|}{\epsilon_s(p)}\right), \quad (16)$$

where $\epsilon_s(p) > 0$ is the fixed metric-depth tolerance. The binary mask $m_s(p)$ equals one only when all tests pass. Among the earlier committed pages, we select the one with the highest score, computed from valid coverage and mean confidence; a target reads at most one page.

If A^0 denotes the native reference-conditioned attention output and A^{hist} the output after reading an eligible page, the historical read is introduced as the bounded residual

$$\begin{aligned} A(p) &= A^0(p) + g_s(p) (A^{\text{hist}}(p) - A^0(p)), \\ g_s(p) &= \lambda_{\text{mem}} m_s(p) \text{clip}(c_s(p), 0, 1), \quad 0 \leq \lambda_{\text{mem}} \leq 1. \end{aligned} \quad (17)$$

Here m_s is the binary geometry-eligibility mask, and c_s combines target/source confidence with metric-depth agreement. When no valid page exists, $g_s = 0$ and the native reference-conditioned path is recovered exactly.

The denoiser exposes clean attention features $Z_{\mathcal{W}_n}^{\text{attn}}$ for selected source latents by re-encoding them at the clean diffusion endpoint. The stored tensor is therefore an attention-value page rather than a decoded RGB frame; the operation changes only the attention features and does not copy pixels, alter the target layout, or mix cache features with simulator visibility fields. The staged page update is

$$\hat{\mathcal{K}}_{n+1} = \mathcal{K}_n \oplus \Phi_{\text{page}}(Z_{\mathcal{W}_n}^{\text{attn}}, \mathcal{G}_n, \Omega_{\mathcal{W}_n}), \quad (18)$$

where $\oplus$ appends pages under their source-frame or source-latent IDs, and $\Omega_{\mathcal{W}_n} = \{\Omega_t : t \in \mathcal{W}_n\}$ is the corresponding reliable static support. Tail evidence is staged from the configured tail RGB frames and remains private, together with the pages, until the quality gate accepts the whole window.

For the evaluated profile, the tail set is $\mathcal{T}_n = \{64, 68, 72, 76, 80\}$. Its candidate evidence is summarized as

$$\hat{\mathcal{E}}_{n+1} = \Phi_{\mathcal{E}}(Y_{\mathcal{W}_n}|_{\mathcal{T}_n}, \mathcal{G}_n, \mathcal{R}), \quad (19)$$

where the operator stores only the aligned generator-depth estimates, trusted support, source indices, and confidence statistics. The simulator-consistent record is passed to this operator for testing purposes only, never for overwriting.

The page and tail branches are both private candidate state. A page created by the current window cannot be selected by another frame within that same window, and tail evidence cannot affect the control tensor already used to sample the current window. Their earliest possible effect occurs in the next accepted window. This ordering renders the memory causal and the depth-completion branch reproducible under retry.

4.6 Quality-Gated Atomic Commit

The quality gate is a fail-closed publication check applied to the decoded window. It examines the media contract together with the fixed boundary, black-frame, and frozen-motion diagnostics; the exact definitions and thresholds are given in the supplementary appendix, Sec. A.5 (“Quality Gate, Atomicity, and Retry Policy”). It is not a replacement for the simulator trajectory metrics.

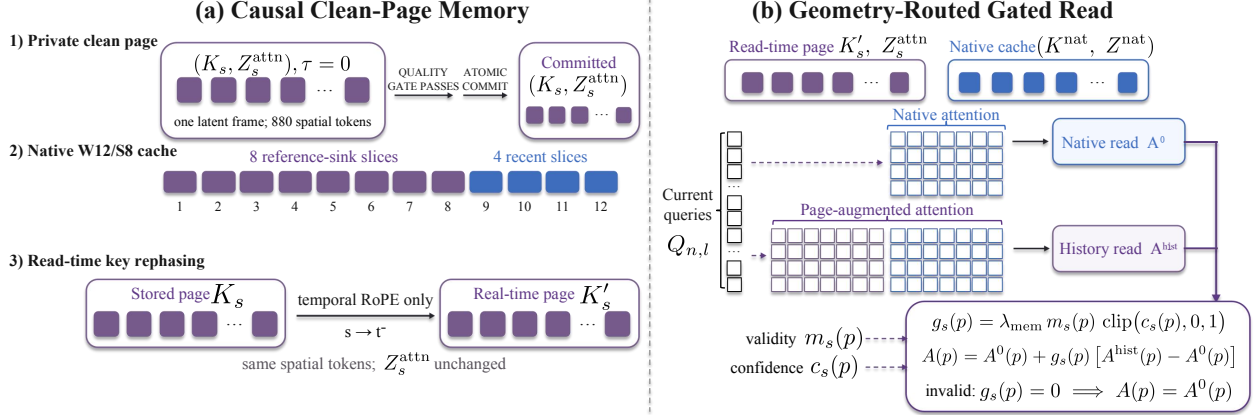


Figure 6 Reference-anchored appearance memory. Historical features become accessible only after their corresponding source window is committed. Simulator-consistent geometry determines whether a target location corresponds to reliable static support in an earlier view. In the diagram, $\mathbf{Z}_s^{\text{attn}}$ denotes the stored attention-value tensor and is distinct from the geometric visibility V_s . The selected historical read is blended with the native reference-conditioned output through a bounded confidence gate. Invalid correspondence recovers the native path exactly.

Let $Y_0, \dots, Y_{80}$ denote the decoded 8-bit RGB frames and B_n the last committed frame. We define $d_t = \delta(Y_t, Y_{t-1})$ for $t > 0$ and compute

$$\begin{aligned} \delta(A, B) &= \frac{1}{3HW} \sum_{p,c} |A(p, c) - B(p, c)|, \\ e_{\text{ov}} &= \delta(Y_0, B_n), \\ r_{\text{bd}} &= \frac{e_{\text{ov}}}{\max\{\text{median}(d_1, \dots, d_8), 10^{-6}\}}. \end{aligned} \tag{20}$$

with $e_{\text{ov}} = 0$ for the first window. The boundary-motion ratio uses the median of the first eight adjacent-frame changes (with a 10^{-6} floor). We also measure the fraction of nearly black pixels and the fraction of adjacent changes below 0.2, denoted r_{black} and r_{freeze} , respectively.

The reported decision is the conjunction

$$\begin{aligned} q_n &= \mathbf{1}[\text{media}] \mathbf{1}[e_{\text{ov}} \leq 4] \mathbf{1}[r_{\text{bd}} \leq 3] \\ &\quad \times \mathbf{1}[r_{\text{black}} \leq 0.10] \mathbf{1}[r_{\text{freeze}} \leq 0.25] \chi_{\text{repr}}. \end{aligned} \tag{21}$$

where the media contract requires 81 decodable 832×480 frames. Here $\chi_{\text{repr}} = 1$ unless a pre-commit feedback-reprojection manifest is supplied, in which case it is the indicator of $e_{\text{repr}} \leq 24$. The manifest was absent in the reported runs. Simulator RGB MAE is logged as a diagnostic and is not an acceptance criterion.

For $q_n = 1$, the terminal physical and camera states, the staged appearance pages, and the staged tail evidence are published together according to Eq. (5). For $q_n = 0$, the working simulator/camera copy, the candidate pages, and the candidate evidence are discarded, and $\mathcal{S}_n$ remains unchanged; the candidate native cache and the random-state checkpoint are restored as well. A retry preserves the same committed index, action, trajectory, generator controls, and committed memory; only the sampling seed may change. The next user action is released only after $q_n = 1$, so a retry cannot advance the physical or epistemic state. Consequently, generated RGB can extend appearance knowledge only through an accepted update and never writes back to X_n , C_n , or D^{geo} .

The gate runs after decoding and before any candidate cache state, page, tail evidence, or terminal simulator state becomes persistent. Thresholds are fixed for the attempt and are never relaxed; an explicit retry reuses the committed checkpoint and all candidate controls, with only the sampling seed permitted to differ. Failed attempts therefore cannot advance either the state index or the history.



Figure 7 Full-duration key-frame comparison across all eight evaluated methods. Within each scene, methods are rows. Each complete baseline output is uniformly mapped to the corresponding 30.06 s TourPhysics duration, and columns sample $t = 5, 15, 25$ s on that matched timeline; no looping, tail padding, or per-output frame selection is used. (a) A pinned curtain under wind. (b) A jelly. Native durations and generation inputs follow Table 1. Frames are uncropped and padded to a common aspect ratio, with no color correction or replacement.

5 Experiments

5.1 Experimental Setup

The experiments are organized around the four capabilities required by the problem formulation. The first experiment asks whether the generated camera and object motion follows the fixed physical and camera trajectory. The second checks whether contacts and deformations occur at the declared times and with the declared shape changes. The third examines whether long-horizon generation preserves the input scene and the intended camera motion. The final experiment isolates the persistent cache and the geometry-routed appearance memory. This order follows Figure 2: the orange physical-transition branch defines the target trajectory, the purple observation branch renders it, and the green lower loop carries visual evidence across revisits.

Benchmark and execution protocol. The benchmark contains 108 simulator-defined interaction sequences initialized from 61 source-image identities. It covers camera-only tours and manipulations of rigid, cloth, elastic, granular, and fluid proxies. Each sequence starts from the same single image and declarative configuration used to initialize the scene hypothesis. For every action, the simulator exports a camera and object trajectory together with the geometric record and a paired simulator rendering; the video model generates

Table 1 Evaluation inputs and temporal support. “Int.” denotes controls derived internally from the declared simulator state, “Text” under Camera denotes prompt-only specification, and ^d marks a uniformly time-stretched full-timeline diagnostic.

Method	Image	Text	Camera	Depth	Track	Mask	Physics script	Native / diagnostic duration
TourPhysics	✓	Scene	Action/pose	Int.	Int.	Int.	Declarative	25.1/30.1 s
LingBot-Cam	✓	Frozen	Pose	–	–	–	–	25.1/30.1 s
LingBot-Act	✓	Frozen	Discrete action	–	–	–	–	25.1/30.1 s
Wan2.1-I2V	✓	Frozen	Text	–	–	–	–	10.7 s / 25–30 s ^d
Sora-2	✓	Frozen	Text	–	–	–	–	12.2 s
MotionCtrl	✓	–	RT pose	–	–	–	–	1.4 s / 25–30 s ^d
Tora-Initiator	✓	Frozen	–	–	Oracle 2D	–	–	6.1 s
minWM-Wan	✓	Frozen	Discrete action	–	–	–	–	39.8 s

Tora receives the simulator-projected initiator trajectory. Depth, track, mask, and physics-script columns are generation-time inputs. The ^d rows use timestamp scaling without looping or tail padding.

an observation for that fixed window. The next action is released only after the observation passes the window-level commit rule. Thus, every paired comparison has a declared target trajectory and an explicit commit boundary, rather than an unconstrained video continuation. The benchmark measures agreement with this controllable hypothesis, which is the quantity needed for exploration and manipulation from one image.

Compared systems and common measurements. The comparison spans the main conditioning regimes used for controllable video and interactive world models. LingBot-Cam and LingBot-Act [49] represent camera- and action-conditioned open-source interactive world-model variants; minWM-Wan [52] represents an interactive video world model. Wan2.1-I2V [28] and Sora-2 [2] are general image-to-video generators whose primary task is appearance and motion synthesis, while MotionCtrl [50] and Tora-Initiator [51] are controllable video generators driven by poses or point trajectories. TourPhysics is the simulator-guided persistent model that receives the declarative scene and computes the physical trajectory before synthesis. Table 1 records the camera, depth, tracking, mask, and physics inputs available to each system, including native and diagnostic durations. This makes the comparison useful for the corresponding task: each method is evaluated through its supported interface, while all outputs are scored against the same simulator-defined targets. Paired object and physical metrics use 8-fps videos at 682×384 and the same frozen-mask CoTracker3 [30, 64, 65] pipeline; camera and appearance protocols are specified with their respective experiments below.

Table 1 summarizes the method-specific generation inputs and native or diagnostic temporal support. It is an input-protocol matrix, rather than a performance table: its purpose is to make the information available to each comparison method explicit before interpreting the results.

5.2 Camera and Object Motion

This experiment tests the most direct consequence of fixing the trajectory before observation synthesis: the camera should execute the prescribed tour and the visible objects should follow the simulator state. It evaluates the camera path and object-motion outputs of the orange “Physical transition” branch in Figure 2, rather than judging motion only by visual smoothness.

Table 2 Prescribed camera- and object-motion adherence. Camera metrics compare generated and declared trajectories; object metrics measure image-space agreement with paired simulator-rendered sequences. Generation inputs are listed in Table 1.

Method	Camera				Object Motion			
	ATE ↓	PLR → 1	T-Dir ↑	R-Dir ↑	ADE ↓	End Err. ↓	S@0.05 ↑	Vis. ↑
LingBot-Cam	0.1031	4.8434	88.54%	53.00%	0.0833	0.1605	18.37%	35.04%
LingBot-Act	0.1235	6.3812	68.75%	61.00%	0.0834	0.1540	27.09%	50.56%
Wan2.1-I2V	0.1436	3.8702	60.42%	42.00%	0.0774	0.1233	31.61%	76.43%
Sora-2	0.0560	2.3898	78.52%	> 99.99% ^{1seg}	0.0721	0.1419	43.47%	81.88%
MotionCtrl	0.0783	0.8517	50.35%	38.00%	0.0871	0.1474	19.74%	88.46%
Tora-Initiator	N/A	N/A	N/A	N/A	0.0462	0.0865	70.05%	80.10%
minWM-Wan	0.1092	1.9820	89.58%	46.00%	0.1051	0.2213	9.46%	69.63%
Ours	0.0233	1.4301	98.26%	93.00%	0.0297[‡]	0.0524[‡]	81.18%[‡]	75.20% [‡]

[‡]TourPhysics object metrics cover 168 objects in 93 sequences from 49 input identities after excluding trajectories with less than 15% joint visibility. Camera metrics use 49 sequences from 40 identities with runtime-exact poses. Sora R-Dir has one eligible segment and is excluded from best-result highlighting. Wan and MotionCtrl are uniformly time-stretched to full duration. Bold marks the best displayed result on each method’s valid support.

Protocol and metrics. Camera scores use the 49 sequences from 40 identities whose reference poses are exact runtime exports. ViPE camera centers are timestamp-matched to the simulator centers, and one Umeyama similarity transform is fitted over each complete sequence. ATE is the RMS camera-center deviation after this alignment, so lower values mean closer spatial tracking. PLR is the recovered-to-declared path-length ratio, with one indicating equal traveled distance. T-Dir and R-Dir are the percentages of translation-only and rotation-only schedule segments whose recovered motion points to the correct half-space. These four metrics separate absolute position error, path scaling, and action-direction correctness.

Object scores use paired simulator and generated videos sampled at 8 fps and 682×384 . Frozen initial masks seed the same CoTracker3 multi-point tracker, and an object is retained only when its centers are jointly visible for at least eight frames and 15% of the sequence. ADE is the mean normalized center error, End Err. is the error at the last jointly visible frame, S@0.05 is the fraction of objects whose ADE is within 5% of the image diagonal, and Vis. is the jointly visible fraction. TourPhysics contributes 168 tracked objects in 93 sequences from 49 identities; the baselines use the valid records listed in the table footnote. Complete alignment and support definitions are provided in the supplementary appendix, Sec. B.2 (“Camera and Object Motion”).

Table 2 shows that TourPhysics has the lowest camera ATE (0.0233) and the highest translation- and rotation-direction hits (98.26% and 93.00%). It also has the lowest object ADE (0.0297) and endpoint error (0.0524), with 81.18% of retained objects within the 5% diagonal threshold. These are the measurements most closely tied to the declared action and state transition: the simulator fixes the path first, and the generated observation preserves that path when it is rendered. PLR and Vis. provide complementary information about total path scaling and track support, rather than replacing the spatial and directional adherence measures.

5.3 Adherence to Simulator-Defined Interactions

The second experiment asks whether a visually plausible interaction also preserves the simulator-defined event sequence. It uses the same paired simulator/generated videos and the orange physical-transition branch in Figure 2, but evaluates event timing, collision response, and deformation rather than whole-object

Table 3 Adherence to simulator-defined physical trajectories. Collision and deformation columns are tracked image-space proxies computed against paired simulator-rendered videos. VideoPhy-2 supplies an auxiliary perceptual diagnostic.

Method	Collision Proxies			Deformation Proxies			VideoPhy-2 AutoEval (Auxiliary)		
	Contact Err. ↓	Response Acc. ↑	Overlap Exc. ↓	Shape Err. ↓	Area Drift ↓	Recovery Acc. ↑	SA ↑	PC ↑	Avg ↑
LingBot-Cam	6.9196 s	< 0.01%	< 10^{-4}	0.0664	0.5136	50.00%	2.9444	4.4167	3.6806
LingBot-Act	7.8304 s	< 0.01%	< 10^{-4}	0.0809	0.1757	< 0.01%	2.9352	4.2315	3.5833
Wan2.1-I2V	5.1875 s	33.33%	< 10^{-4}	0.0866	0.2763	30.00%	2.9444	4.0185	3.4815
Sora-2	4.9107 s	33.33%	< 10^{-4}	0.1351	0.3369	33.33%	2.9630	4.2778	3.6204
MotionCtrl	7.0893 s	< 0.01%	< 10^{-4}	0.1234	0.3539	N/A	2.9352	4.7963	3.8657
Tora-Initiator	5.8839 s	< 0.01%	< 10^{-4}	0.1081	0.2249	> 99.99% ^{1evt}	2.8660	4.2784	3.5722
minWM-Wan	4.5639 s	22.92%	< 10^{-4}	0.1530	0.5046	< 0.01%	2.9352	4.1296	3.5324
Ours	3.3304 s ^C	16.67% ^C	< 10^{-4} ^C	0.0704 ^M	0.0866 ^M	64.29% ^R	2.9167	4.3148	3.6157

“N/A” denotes an incompatible protocol. ^CContact, response, and overlap cover 14/5/9 valid TourPhysics sequences from 6/3/5 input identities; ^Mshape and area cover 30 records in 25 sequences from 13 identities; ^Rrecovery covers eight events in seven sequences from two identities. Tora recovery has one eligible event and is excluded from best-result highlighting. All deterministic columns use the same frozen CoTracker3 pipeline, with method-specific support after visibility filtering; bold marks the best displayed value on that method-specific support rather than on a common subset. VideoPhy-2 covers 108 sequences (Tora: 97) and reports custom-data mean SA/PC on a 1–5 scale; Avg is their arithmetic mean. Wan and MotionCtrl are time-stretched to full duration. Zero displayed 2D overlap does not imply zero 3D penetration.

centers.

Protocol and metrics. The simulator supplies the contact onset, initiator-target pair, and declared deformation interval for each sequence. The generated and simulator videos are tracked with the frozen-mask CoTracker3 protocol at 8 fps; records are reduced within a sequence and then averaged equally over sequences, so a sequence with several objects or events does not dominate. Contact Err. is the timing error between the generated and simulator contact proxies. Response Acc. is the percentage of observable targets that exhibit the prescribed post-contact motion. Overlap Exc. is the excess 90th-percentile 2D hull overlap after contact. Shape Err. is the normalized Procrustes point error for a deformable object, Area Drift is the deviation of its projected area from the initial ratio, and Recovery Acc. is the percentage of declared deformation events that meet both the amplitude and post-event recovery criteria. The auxiliary SA and PC scores from VideoPhy-2 [53] measure semantic adherence and physical commonsense on a 1–5 scale; their average is a perceptual diagnostic, not a trajectory score. The event definitions and effective support for each method are given in the supplementary appendix, Sec. B.3 (“Simulation-Trajectory Adherence”).

Table 3 shows the strengths of the simulator-guided branch on the deterministic interaction measures. TourPhysics reaches the lowest contact-timing error (3.3304 s), the lowest area drift (0.0866), and the highest recovery accuracy (64.29%); its shape error (0.0704) is second only to LingBot-Cam on that supported subset. The contact and deformation columns answer different parts of the same question: the first group tests when an event happens, while the second tests whether the generated object changes in the right way and returns toward its post-event state. The perceptual VideoPhy-2 columns are reported alongside these deterministic measures as an auxiliary appearance diagnostic; the deterministic columns provide the trajectory result.

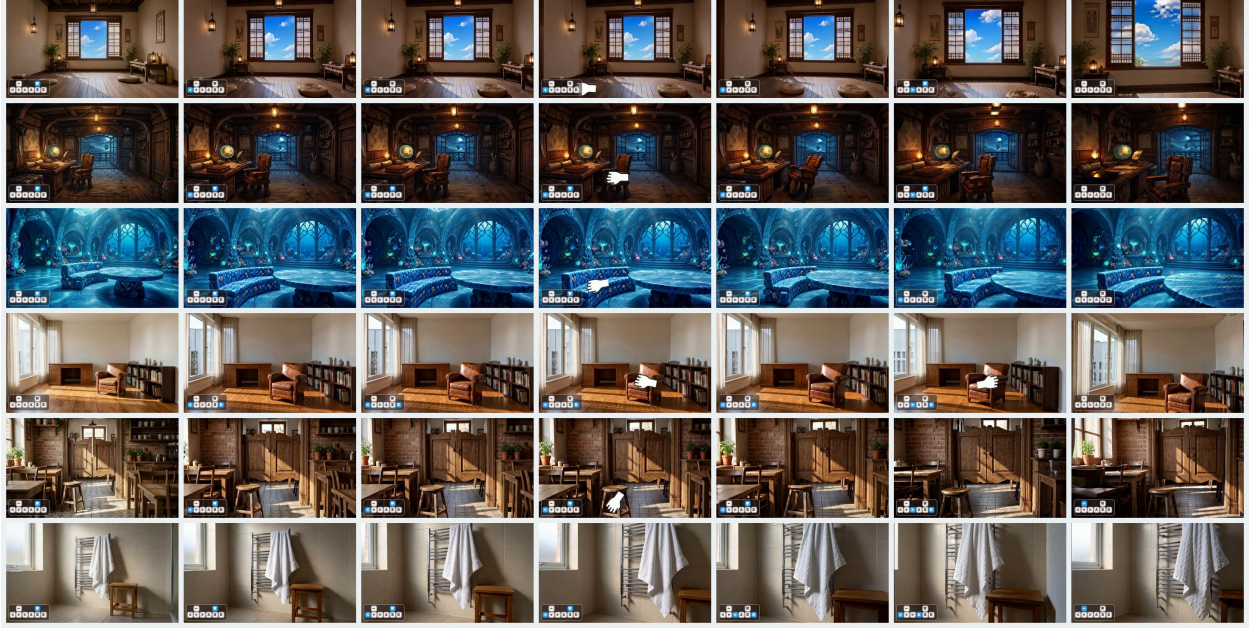


Figure 8 TourPhysics results across several scenarios. Key frames progress from left to right and show long-horizon appearance, camera motion, and physical interactions.

5.4 Appearance and Camera-Control Diagnostics

The trajectory experiments establish control, while this experiment checks whether that control is accompanied by stable visual content. Figure 7 uses all eight systems on two representative scenes, a pinned curtain under wind and a jelly. Each complete baseline output is mapped to the 30.06-s TourPhysics timeline and sampled at 5, 15, and 25 s, with no looping, tail padding, or per-output frame selection. Figure 8 extends the view to several long-horizon camera and interaction scenarios.

The quantitative diagnostics measure complementary properties of the generated observations. Input-video subject consistency measures whether the foreground appearance remains aligned with the source video, background consistency tests the static scene, and camera-motion classification accuracy measures whether the generated motion retains the intended camera-motion class. TourPhysics obtains 0.9743, 0.9791, and 0.8921 on these three quantities, respectively. These diagnostics connect the purple observation branch and the green memory loop in Figure 2 to the visible long-horizon behavior: the model can follow a physical/-camera trajectory while retaining the reference scene. The underlying masks, aggregation, and temporal normalization are specified in the supplementary appendix, Sec. B.4 (“Appearance and Long-Horizon Consistency”).

5.5 Long-Horizon Cache and Appearance Memory

The final experiment targets the failure mode that motivates persistent memory: a camera can revisit a surface after many intervening actions, and a finite temporal cache alone does not preserve its appearance. It isolates the green lower loop and long-horizon filmstrip in Figure 2 in two stages. The cache comparison uses 72 paired long videos from 27 identities and keeps the observation backbone and generator fixed while replacing an ordinary finite cache with the persistent sink cache. The memory comparison uses four development identities and an identity-disjoint held-out set of 14 dynamic identities. Each selected sequence has six windows, 481 output frames, 51 planned target latents, and revisit chunks 3–5; physical and camera trajectories, generator-facing depth, checkpoints, prompts, and random seeds are matched between memory-on and memory-off runs.

The cache metrics are end-to-start drift and tail-to-head drift, both computed from the mean decoded 8-bit channel value, so lower values mean less long-range brightness change. The memory metrics are relative

Table 4 Long-horizon observation and appearance-memory diagnostics. Cache drift is measured on paired long-video sequences. Memory changes are reported relative to the same generator with appearance memory disabled; negative values denote lower error.

Variant	Scope	End-to-start drift ↓	Tail-to-head drift ↓	History error change ↓	Known-region change ↓
Ordinary temporal cache	72 sequences	26.33	20.38	–	–
Persistent sink cache	72 sequences	11.17	7.74	–	–
Appearance memory, development	4 identities	–	–	–8.25%	–2.99%
Appearance memory, held-out	14 identities	–	–	–2.39%	–0.90%

changes in historical-region error on routed, depth-consistent correspondences and known-region error on the simulator-consistent background; negative percentages indicate lower error with memory. This separation follows the distinction between temporal association and appearance filling emphasized by memory-based video segmentation and temporally-aware inpainting [62, 63]. Formula definitions, masks, aggregation, and the identity-disjoint selection protocol are provided in the supplementary appendix, Sec. B.4 (“Appearance and Long-Horizon Consistency”).

Table 4 separates the two mechanisms cleanly. The persistent sink cache reduces end-to-start drift from 26.33 to 11.17 and tail-to-head drift from 20.38 to 7.74. Geometry-routed appearance memory then reduces historical-region error by 8.25% on development and 2.39% on the held-out identities, while known-region error changes by -2.99% and -0.90% . The first comparison shows why the cache is needed for long windows; the second shows that accepted, geometry-routed history improves revisits while leaving the physical and camera trajectory and generator-facing depth unchanged.

6 Conclusion

We presented TourPhysics, an online framework for exploring and manipulating a controllable scene hypothesis initialized from a single image and a declarative physical configuration. Its architecture assigns separate roles to simulator state, simulator-consistent geometry, generator controls, and accepted appearance memory.

For each action, TourPhysics computes the physical and camera trajectory before generating an observation. Accepted observations publish the terminal state and update later appearance memory and generator-facing depth, while the committed state and geometry remain fixed during synthesis and retry. A permanent image reference and geometry-routed memory preserve appearance when the camera revisits earlier surfaces.

On simulator-defined camera tours and manipulations, TourPhysics improves alignment with prescribed camera and object trajectories, preserves the input scene, and reduces long-horizon appearance drift. Future work will study real-world physical evaluation, more complete scene geometry, and fixed-capacity memory for longer tours.

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A Implementation and Evaluation Details

This appendix summarizes implementation and evaluation details omitted from the main paper. The single-image decomposition, explicit scene alignment, and multi-physics initialization build on the corresponding components of the ACM Multimedia 2026 conference paper PhysOmni [54]. TourPhysics retains this physical front end but extends it with persistent online state, action-boundary trajectory simulation, transactional observation publication, separated geometric and generative controls, and long-horizon appearance memory.

A.1 Single-Image Scene Construction

Instance decomposition and explicit proxies. Given the input image I_0 and user-provided object descriptions, SAM 3 [55] extracts instance masks $\{M_i\}_{i=1}^N$. Each image-mask pair is then lifted by SAM 3D Objects [56] into an explicit triangular mesh $\mathcal{S}_i^{\text{loc}}$ and an object-to-camera similarity transform (s_i, R_i, t_i) . A local vertex is placed in the common camera frame by

$$v_i^{\mathcal{C}} = s_i R_i v_i^{\text{loc}} + t_i. \quad (\text{A1})$$

These explicit surfaces provide the simulator’s contact and collision interfaces. The foreground union is removed cumulatively with LaMa inpainting [57, 62, 77] to form a clean static background plate,

$$B_0 = \mathcal{F}_{\text{inp}} \left(I_0, \bigcup_{i=1}^N M_i \right). \quad (\text{A2})$$

The unmodified I_0 is retained separately as the permanent appearance reference. The inpainted plate is used only for camera-induced background exposure.

Gravity-aligned scene coordinates. Independently reconstructed meshes share an image-conditioned camera frame but need a stable gravity direction and ground height before simulation. Let $\hat{u}_{\mathcal{C}}$ be the provisional up direction. For each mesh, we select the lowest η -quantile of vertices under $h(v) = v^{\top} \hat{u}_{\mathcal{C}}$, and pool these likely contact anchors into $\mathcal{P}_{\text{base}}$. We use $\eta = 0.05$ by default and relax it to 0.10 when too few anchors are available. A robust plane is then fitted as

$$(n^*, d^*) = \arg \min_{\|n\|_2=1, d} \sum_{p \in \mathcal{P}_{\text{base}}} \rho(n^{\top} p + d), \quad (\text{A3})$$

where ρ is implemented by RANSAC inlier selection. Restricting the fit to low object anchors prevents dense vertical structures from dominating the estimated ground. A rigid transform rotates n^* onto the world z -axis, recenters the pooled scene, and translates the lowest transformed anchor to $z = 0$. Small residual pairwise overlaps are corrected by bounded translations along the minimum-overlap bounding-box axis. These corrections are part of the declared initial hypothesis, not online simulator updates.

Reference-camera and depth alignment. Monocular depth and camera estimates provide the initial reference-view geometry. We refine the camera by rendering the union silhouette of the aligned meshes and maximizing its overlap with $\bigcup_i M_i$. A coarse-to-fine coordinate search translates the camera and look-at point together along the forward, right, and up axes, followed by a bounded image-plane correlation refinement. For visible pixels with valid monocular and simulator depth, we fit an affine alignment using the same scale and shift (α, β) as in the main text:

$$D_0^{\text{geo}}(p) = \alpha \tilde{D}_0(p) + \beta, \quad (\text{A4})$$

where D_0^{geo} denotes simulator-consistent depth. Invalid or non-finite pixels are excluded from the fit. This alignment establishes the internal scene coordinates up to the scale supported by the single-image hypothesis.

The source image and its initial geometric record are stored permanently:

$$\mathcal{R} = (I_0, B_0, D_0^{\text{geo}}, V_0, Q_0, L_0, K_0, T_0). \quad (\text{A5})$$

Here, B_0 is the cached inpainted background plate; V_0 , Q_0 , and L_0 denote validity, confidence, and object labels, while K_0 and T_0 are camera intrinsics and world-to-camera extrinsics. The reference image and its geometry remain read-only during online interaction.

A.2 Declarative Configuration and Physical Simulation

Configuration proposal, validation, and override. TourPhysics accepts a directly authored physical declaration. As an optional initializer, Qwen2.5-VL-72B-Instruct [78] receives the complete image and the masked object crops and proposes a structured configuration. For object i , the record contains a material family m_i , continuous parameters θ_i , a boundary condition b_i , and an optional initial velocity; the scene record additionally contains velocity keyframes, scheduled force fields, and action mappings:

$$\Theta_{\text{phys}} = (\{m_i, \theta_i, b_i, v_i^0\}_{i=1}^N, \mathcal{U}, \mathcal{F}). \quad (\text{A6})$$

The supported families cover rigid bodies; MPM elastic, elastoplastic, granular, snow, and liquid materials; and PBD elastic, cloth, liquid, and particle materials. Their parameters include density, friction, Young’s modulus, Poisson ratio, yield stress or angle, compliance, relaxation, and air resistance as applicable. Boundary conditions include fixed objects, ground support, and particle pinning; for example, a declared top fraction of a cloth mesh can be pinned to represent a hanging attachment. Supported external fields include constant, wind, point, drag, vortex, turbulence, and noise fields, each with an activation schedule.

The proposal is parsed against a material and force-field whitelist, completed with family-specific defaults, and exposed for manual editing. The resulting declaration is frozen before the first trajectory simulation, so the VLM participates only in initialization. This frozen record is Θ_{decl} in the main paper.

Multi-physics solver routing. We use the GPU-accelerated Genesis engine [79]. Non-deformable objects and articulated agents are routed to rigid-body dynamics (RBD). In generalized coordinates, the update solves [80]

$$M(q)\ddot{q} + h(q, \dot{q}) = \tau + J(q)^\top f_{\text{ext}}, \quad (\text{A7})$$

where M is the inertia matrix, h collects velocity-dependent and gravitational terms, and $J^\top f_{\text{ext}}$ maps contact and external forces into generalized coordinates.

Continuum materials and large volumetric deformations are handled with the Material Point Method (MPM) [27]. Material state is carried by Lagrangian particles and momentum is updated on an Eulerian grid. In particular, the internal force accumulated at grid node j is

$$f_j^{\text{int}} = - \sum_p V_p \sigma_p \nabla w_{jp}, \quad (\text{A8})$$

where V_p and σ_p are particle volume and stress, and w_{jp} is the particle–grid interpolation kernel. Particle-to-grid transfer, the grid momentum update, and grid-to-particle transfer naturally support elastic, plastic, granular, and fluid trajectories.

Thin shells and constraint-dominated particle systems are handled with Position-Based Dynamics (PBD) [3]. For a constraint $C(x) = 0$, the position correction for particle i is

$$\Delta x_i = -w_i \frac{C(x)}{\sum_j w_j \|\nabla_{x_j} C(x)\|_2^2 + \epsilon} \nabla_{x_i} C(x), \quad (\text{A9})$$

where w_i is inverse mass. Iterative projection enforces stretch, bending, volume, density, and attachment constraints.

RBD–MPM and RBD–PBD contacts are enabled through the engine coupler. The solvers advance within the same scene step, exchange contact response at their interfaces, and retain solver-native state for the next committed window. Static support uses a rigid plane placed at the aligned scene floor. Unless a scene declaration overrides them, the solver step is 4 ms, with 10 substeps for rigid-only scenes and 16 for scenes containing particle materials; gravity is $(0, 0, -9.8)$. The default MPM and PBD domains are $[-3, 3]^3$; MPM uses particle size 0.02 and grid density 32, whereas PBD uses particle size 0.01. These values define the controllable scene hypothesis.

Authoritative trajectory simulation and observation sampling. For every online action $a_n = (c_n, u_n)$, the simulator and camera controller are advanced on a working copy of the preceding committed checkpoint before observation synthesis:

$$(\mathbf{X}_n, \mathbf{C}_n, \mathcal{G}_n) = \mathcal{F}_{\text{seq}}(\tilde{X}_n, \tilde{C}_n, a_n; \Theta_{\text{decl}}), \quad (\text{A10})$$

The resulting trajectory segment records the simulator state, camera pose, simulator-consistent depth, visibility, object labels, and geometric confidence at every observation frame. This segment remains fixed during video generation, and its terminal state is committed only after the resulting observation passes the quality gate. At each source frame, the simulator renderer exports metric depth, entity-ID segmentation, camera intrinsics and extrinsics, and the dynamic-object state used to derive visibility and motion controls. These rendered buffers populate $\mathcal{G}_n$ and condition observation synthesis. The video model’s decoder supplies the output RGB.

The simulator control record is indexed at $f_{\text{sim}} = 60$ fps, while generated observations use $f_{\text{gen}} = 16$ fps. Let o_n be the sampling origin associated with window n , rather than the committed terminal state index. To avoid floating-point timestamp drift, generated frame j selects simulator frame

$$k_n(j) = o_n + \left\lfloor \frac{(2j+1)f_{\text{sim}}}{2f_{\text{gen}}} \right\rfloor. \quad (\text{A11})$$

The initial origin o_0 follows the simulator’s frame-index convention. Set $o_{n+1} = o_n + 300$, so $k_{n+1}(0) = k_n(80)$. Thus, under this center-of-bin convention, the shared boundary is the sampled frame $k_n(80)$; the committed terminal states X_n^+, C_n^+ are defined at that endpoint, not at the raw origin $o_n + 300$. Consecutive observations contain 81 frames and share one simulator/conditioning boundary, corresponding to an 80-frame stride; their decoded RGB boundary is checked by the overlap gate and deduplicated when windows are concatenated. The next action becomes available only after the current window is committed. The simulated trajectory, including all intervention and force-field activation indices, remains unchanged during generation and retry.

A.3 Separated Depth Controls

Let n index committed interaction windows and $t \in \mathcal{W}_n$ index frames within a window; retry attempts retain the same n . TourPhysics maintains the following per-frame quantities:

$$D_{n,t}^{\text{geo}}, \quad D_{n,t}^{\text{gen}} = [\mathcal{D}_{\mathcal{W}_n}^{\text{gen}}]_t, \quad D_{n,t}^{\text{base}} = [\mathcal{D}_{\mathcal{W}_n}^{\text{base}}]_t. \quad (\text{A12})$$

Simulator-consistent depth $D_{n,t}^{\text{geo}}$ is metric camera-space z -depth in calibrated scene units; T_t is world-to-camera and $\pi^{-1}(p, z; K_t)$ returns a camera-frame point with third coordinate z . It is used for projection, visibility, object correspondence, and evaluation. Scene initialization and simulator rendering are its only writers. The base window tensor $\mathcal{D}_{\mathcal{W}_n}^{\text{base}} = \mathcal{F}_{\text{base}}(\mathcal{G}_n; \mathcal{R})$ is a normalized relative-depth condition supplied to the video model.

The persistent state stores accepted tail evidence $\mathcal{E}_n$, not a writable depth map. At the beginning of the current window, after $\mathcal{G}_n$ has been fixed, the generator-facing tensor is constructed as

$$\mathcal{D}_{\mathcal{W}_n}^{\text{gen}} = \mathcal{F}_{\text{ctrl}}(\mathcal{G}_n, \mathcal{E}_n) = (1 - W_n) \odot \mathcal{D}_{\mathcal{W}_n}^{\text{base}} + W_n \odot \overline{\mathcal{D}}_{\mathcal{W}_n}(\mathcal{E}_n), \quad (\text{A13})$$

where $W_n \in [0, 1]^{|\mathcal{W}_n| \times H \times W}$ selects eligible newly exposed regions inadequately covered by the current generator-facing base control. The candidate completion is estimated only from committed tail evidence and is applied only when novelty, overlap, temporal consistency, and the affine-aligned depth-agreement tests are satisfied. The reported gates are trusted fraction ≥ 0.20 , Spearman ≥ 0.70 , model-space NRMSE ≤ 0.08 , and metric-space NRMSE ≤ 0.12 , with novelty, target-hole, and source-support fractions 0.05, 0.01, 0.20, respectively. These tests are pointwise after the frame-level admissibility checks; otherwise $W_n(t, p) = 0$ and the base control is retained exactly at that location.

Generated depth affects only visual conditioning; collision geometry, object state, and simulator-consistent depth remain unchanged.

A.4 Observation Generation and Transactional State

The observation model is a finite-window video diffusion model trained for geometry-conditioned observation synthesis:

$$Y_{\mathcal{W}_n} \sim p_{\theta}(Y_{\mathcal{W}_n} \mid \mathcal{G}_n, \mathcal{D}_{\mathcal{W}_n}^{\text{gen}}, I_0, \mathcal{K}_n). \quad (\text{A14})$$

The physical and camera state sequence and its geometric controls remain fixed while $Y_{\mathcal{W}_n}$ is generated. The native temporal cache is used for short-range continuity, whereas the appearance memory provides sparse long-range access to previously accepted static appearance.

For a configured tail subset $\mathcal{T}_n$ (the reported profile uses $\{64, 68, 72, 76, 80\}$), the candidate evidence is

$$\hat{\mathcal{E}}_{n+1} = \Phi_{\mathcal{E}}(Y_{\mathcal{W}_n} | \mathcal{T}_n, \mathcal{G}_n, \mathcal{R}), \quad (\text{A15})$$

and remains private until the quality decision accepts the full window. The operator stores only generated-depth estimates, trusted/static support, indices, and alignment/confidence statistics; simulator-consistent geometry is never overwritten.

The corresponding appearance pages are staged from clean attention features and reliable static support:

$$\hat{\mathcal{K}}_{n+1} = \mathcal{K}_n \oplus \Phi_{\text{page}}(Z_{\mathcal{W}_n}^{\text{attn}}, \mathcal{G}_n, \Omega_{\mathcal{W}_n}), \quad (\text{A16})$$

where $\Omega_{\mathcal{W}_n} = \{\Omega_t : t \in \mathcal{W}_n\}$ is the reliable static support defined below. The pages remain private, together with the candidate evidence, until the enclosing window commits.

Candidate updates remain private during generation. If the quality decision is $q_n = 1$, the state is committed as

$$\mathcal{S}_{n+1} = (X_n^+, C_n^+, \hat{\mathcal{K}}_{n+1}, \hat{\mathcal{E}}_{n+1}). \quad (\text{A17})$$

If $q_n = 0$, the persistent state remains unchanged and the working simulator and camera copy is discarded:

$$\mathcal{S}_{n+1} = \mathcal{S}_n. \quad (\text{A18})$$

In this rejection equation the $n+1$ label is only a bookkeeping view; the committed index does not advance. The simulator state, memory pages, and depth evidence remain at the preceding committed boundary; a retry keeps the same committed index and uses the same action, trajectory, controls, and committed memory. If $q_n = 1$, $\hat{\mathcal{E}}_{n+1}$ becomes the committed $\mathcal{E}_{n+1}$. After the next action fixes $\mathcal{G}_{n+1}$, Eq. (A13) constructs $\mathcal{D}_{\mathcal{W}_{n+1}}^{\text{gen}}$; no candidate evidence can be used before that commit.

A.5 Quality Gate, Atomicity, and Retry Policy

The quality gate is a fail-closed publication check, not an evaluation metric for physical fidelity. It runs after all 81 candidate RGB frames have been rendered and decoded, but before the candidate cache state, appearance pages, tail evidence, or terminal simulator state becomes persistent. The media contract requires exactly 81 decodable frames at 832×480 . A contract violation rejects the candidate before any state publication.

Let $Y_0, \dots, Y_{80}$ be the decoded candidate frames on the 8-bit RGB scale, and let B_n be the last frame of the preceding committed window. We use the full-frame mean absolute RGB difference

$$\delta(A, B) = \frac{1}{3HW} \sum_{p,c} |A(p, c) - B(p, c)| \quad (\text{A19})$$

and adjacent-frame changes $d_t = \delta(Y_t, Y_{t-1})$, $t = 1, \dots, 80$. For windows after the first, the shared-boundary error and boundary-motion ratio are

$$e_{\text{ov}} = \delta(Y_0, B_n), \quad r_{\text{bd}} = \frac{e_{\text{ov}}}{\max\{\text{median}\{d_1, \dots, d_8\}, 10^{-6}\}}. \quad (\text{A20})$$

The first window has no preceding boundary and sets both quantities to zero. We also compute

$$r_{\text{black}} = \frac{1}{81HW} \sum_{t,p} \mathbf{1} \left[\max_c Y_t(p, c) \leq 3 \right], \quad (\text{A21})$$

$$r_{\text{freeze}} = \frac{1}{80} \sum_{t=1}^{80} \mathbf{1}[d_t < 0.2]. \quad (\text{A22})$$

Thus, black pixels and frozen transitions are defined directly on decoded 8-bit frames rather than by a learned quality model.

For the reported runs, the binary decision is the conjunction

$$q_n = \mathbf{1}[\text{media contract}] \mathbf{1}[e_{\text{ov}} \leq 4.0] \mathbf{1}[r_{\text{bd}} \leq 3.0] \cdot \mathbf{1}[r_{\text{black}} \leq 0.10] \mathbf{1}[r_{\text{freeze}} \leq 0.25] \chi_{\text{repr}}, \quad (\text{A23})$$

where $\chi_{\text{repr}} = \mathbf{1}[e_{\text{repr}} \leq 24.0]$ when a pre-commit feedback-reprojection manifest is available, and $\chi_{\text{repr}} = 1$ otherwise. No manifest was supplied in the reported video runs, so the optional field was recorded as zero and did not vary their acceptance decisions. Known-background and dynamic-foreground RGB MAE against the paired simulator render are also logged on the 0–255 scale. They are diagnostics only: simulator RGB is a conditioning reference rather than a pixel target, so these two MAEs do not enter Eq. (A23). Thresholds are fixed for an attempt and are never relaxed after a failure.

If $q_n = 0$, the staged video/cache state and candidate evidence are discarded, the working simulator/-camera copy is released, and the rejection rule together with the main-paper knowledge-transition rule leave every persistent component at the last committed boundary. The implementation performs no automatic retry. An explicit retry starts from the same checkpoint with the same committed index, action, trajectory, controls, and committed memory, although the caller may supply a different sampling seed. A later user action becomes available after a candidate passes the unchanged gate and commits.

A.6 Reference-Anchored Appearance Memory

Only reliable static regions can enter long-term memory. The eligible support is defined as

$$\Omega_t = V_t \cap M_t^{\text{sta}} \cap \{Q_t \geq \tau_q\} \cap \neg \Gamma_t^{\text{edge}}, \quad (\text{A24})$$

where M_t^{sta} is derived from the declared static support and the motion/deformation field, Γ_t^{edge} excludes unreliable boundaries, and τ_q is a fixed confidence threshold. Moving and deforming objects are excluded from long-term memory. The K_t and T_t symbols below denote the camera-intrinsic and world-to-camera fields of the corresponding geometric record. The page extraction and private publication rule follow Eq. (A16); no page is readable before the enclosing window commits.

For a target pixel p , simulator-consistent depth back-projects the pixel into the shared scene coordinate system as metric camera-space z -depth, with T_t world-to-camera:

$$x_w = T_t^{-1} \pi^{-1}(p, D_t^{\text{geo}}(p); K_t). \quad (\text{A25})$$

Projection into an earlier committed view s gives

$$(\hat{p}_s, \hat{z}_s) = \pi(T_s x_w; K_s). \quad (\text{A26})$$

A source view is eligible only when the projected location is valid, visible, static, and depth-consistent. The correspondence confidence is

$$c_s(p) = Q_t(p) Q_s(\hat{p}_s) \exp\left(-\frac{|D_s^{\text{geo}}(\hat{p}_s) - \hat{z}_s|}{\epsilon_s(p)}\right). \quad (\text{A27})$$

Here $\epsilon_s(p) > 0$ is the metric-depth uncertainty tolerance, and the binary correspondence mask $m_s(p)$ is one exactly when the projected source passes the in-bounds, visibility, static-support, and depth-consistency tests.

Let A^0 denote the native reference-conditioned attention output and A^{hist} the output after adding an eligible historical page. Historical appearance is introduced only as a bounded residual:

$$A(p) = A^0(p) + g_s(p) (A^{\text{hist}}(p) - A^0(p)), \quad (\text{A28})$$

where

$$g_s(p) = \lambda_{\text{mem}} m_s(p) \text{clip}(c_s(p), 0, 1). \quad (\text{A29})$$

If no valid source view exists, $m_s(p) = 0$, and the output exactly recovers the native path $A(p) = A^0(p)$.

A candidate memory page is stored in a private staging area while its observation is evaluated. It becomes readable only after the observation window commits. This rule prevents failed or rejected generations from contaminating future observations.

B Additional Evaluation Protocol

B.1 Benchmark and Comparison Setting

The evaluation contains simulator-defined camera tours and object manipulations involving rigid, cloth, elastic, granular, and fluid proxies. The benchmark is initialized from 61 source-image identities and contains 108 interaction sequences. When several sequences share an input image, we report both sequence-level and identity-balanced statistics where appropriate.

The benchmark evaluates agreement with a declared simulation trajectory. It does not test whether the chosen declarative parameters are the true physical parameters of the source scene. This distinction is important because single-image initialization does not uniquely determine scale, occluded geometry, density, friction, or deformation parameters.

Compared methods expose different conditioning interfaces. Some methods receive only an image, while others receive an explicit camera trajectory, event description, or privileged control. Such differences are indicated in the corresponding tables. Methods producing shorter videos may be temporally normalized for generic video-quality diagnostics, but these results are not treated as equivalent to native online generation.

B.2 Camera and Object Motion

Camera coordinates and alignment. We retain only the 49 sequences whose reference poses are exact runtime exports. For sequence s , ViPE recovers camera centers $\hat{c}_{s,t}$, which are timestamp-matched to reference centers $c_{s,t}$ in the internal simulator world coordinate system. One Umeyama similarity transform (γ_s, R_s, u_s) is fitted over every valid frame of the complete sequence, never separately by window or segment:

$$\tilde{c}_{s,t} = \gamma_s R_s \hat{c}_{s,t} + u_s. \quad (\text{B1})$$

If $\mathcal{V}_s$ is the valid timestamp set, the per-sequence absolute trajectory error (ATE) is

$$\text{ATE}_s = \sqrt{\frac{1}{|\mathcal{V}_s|} \sum_{t \in \mathcal{V}_s} \|\tilde{c}_{s,t} - c_{s,t}\|_2^2}. \quad (\text{B2})$$

ATE is expressed in simulator world units after alignment; those units belong to the declared scene hypothesis and are not recovered real-world meters. The path-length ratio (PLR) uses the same valid frames and transform:

$$\text{PLR}_s = \frac{\sum_t \|\tilde{c}_{s,t} - \tilde{c}_{s,t-1}\|_2}{\sum_t \|c_{s,t} - c_{s,t-1}\|_2}. \quad (\text{B3})$$

It is dimensionless and has ideal value one because the recovered and reference cameras then travel equal total distance. The main-paper camera/object-motion table reports equal-weight means over valid sequences for both ATE and PLR.

T-Dir and R-Dir use schedule segments that are fully covered, last at least 0.5 s, and contain exclusively translation or rotation commands, respectively. For each segment we form the recovered and reference relative transforms between its endpoints. Let v^{pred} and v^{ref} denote their translation vectors for T-Dir or their axis-angle rotation vectors for R-Dir. A direction hit is

$$h = \mathbf{1} \left[\frac{v^{\text{pred}} \cdot v^{\text{ref}}}{\|v^{\text{pred}}\|_2 \|v^{\text{ref}}\|_2} > 0 \right]. \quad (\text{B4})$$

The test therefore checks the correct half-space, not exact-axis recovery. Hits are averaged first over eligible segments in a sequence and then equally over sequences; the table reports percentages.

Object coordinates and aggregation. The paired simulator and generated videos are sampled at 8 fps and 682×384 . Frozen initial object masks seed the same CoTracker3 multi-point procedure in both videos, and a robust center is computed for each dynamic object. With $D = \sqrt{682^2 + 384^2}$, reference/generated centers $x_{s,i,t}$, $\hat{x}_{s,i,t}$, and joint-visibility indicator $v_{s,i,t}$, the normalized error is

$$e_{s,i,t} = \frac{\|\hat{x}_{s,i,t} - x_{s,i,t}\|_2}{D}. \quad (\text{B5})$$

Table 5 Effective support for the camera- and object-motion protocols. Each cell lists counts in the order specified in the table note; “—” denotes an incompatible protocol.

Method	Camera			Object Motion
	ATE/PLR	T-Dir	R-Dir	Object Motion
LingBot-Cam	49/40	70/48/40	52/25/25	74/49/27
LingBot-Act	49/40	70/48/40	52/25/25	97/51/29
Wan2.1-I2V	49/40	70/48/40	52/25/25	165/92/49
Sora-2	49/40	57/45/37	1/1/1	186/97/53
MotionCtrl	49/40	70/48/40	52/25/25	179/95/51
Tora-Initiator	—	—	—	175/96/52
minWM-Wan	49/40	70/48/40	52/25/25	177/95/51
TourPhysics	49/40	70/48/40	52/25/25	168/93/49

ATE/PLR counts are sequences/identities; T-Dir and R-Dir counts are segments/sequences/identities; object-motion counts are objects/sequences/identities. All counts are after method-specific visibility filtering.

Thus, the 0.05 threshold is five percent of the image diagonal in normalized image coordinates. We retain a dynamic object only if at least eight frames have jointly visible robust centers and its joint-visibility fraction is at least 0.15. For a retained object, with $N_{s,i} = \sum_t v_{s,i,t}$, and last jointly visible frame $t_{s,i}^*$,

$$\text{ADE}_{s,i} = \frac{1}{N_{s,i}} \sum_t v_{s,i,t} e_{s,i,t}, \quad (\text{B6})$$

$$\text{End}_{s,i} = e_{s,i,t_{s,i}^*}, \quad (\text{B7})$$

$$\text{S@0.05}_{s,i} = \mathbf{1}[\text{ADE}_{s,i} \leq 0.05], \quad (\text{B8})$$

$$\text{Vis}_{s,i} = N_{s,i}/T_s. \quad (\text{B9})$$

Each quantity is averaged over retained objects within a sequence and then equally over sequences. ADE, End Err., S@0.05, and Vis. are dimensionless and share the same method-specific support.

Effective support. Table 5 reports the method-specific support after visibility filtering. The table note states the unit and ordering of each count; identities are distinct input-image SHA-256 values.

B.3 Simulation-Trajectory Adherence

The six deterministic metrics use the same frozen-mask, multi-point CoTracker3 observations from paired simulator/generated RGB at 8 fps and 682×384 . All distances below are divided by the image diagonal $D = \sqrt{682^2 + 384^2}$. A record is first reduced within its sequence; the reported value is then an equal-weight mean over sequences. Consequently, sequences with several objects or events do not receive extra weight.

Contact Err. For initiator point set P_k and target point set Q_k , the separation curve is

$$g_k = \min_{p \in P_k, q \in Q_k} \|p - q\|_2 / D. \quad (\text{B10})$$

We median-filter g_k over a three-frame neighborhood and search from the declared onset through the next 8 s. At least eight valid samples and an approach amplitude of 0.002 diagonal are required. The initial level is the maximum in the early part of the search window (up to 16 samples), and the minimum level is its fifth percentile. The contact proxy is the first sample below the minimum plus 15% of the initial-to-minimum range, or the closest approach if no sample crosses that level. The simulator selects the earliest valid target,

and the generated video is scored against that same target. If the generated target is observable but has no valid approach, its error is the 8 s search-window penalty. Evaluation frame k is converted by $t_k = k/8$ seconds (equivalently, by its stored 8-fps timestamp), and the reported record is $|t_k^{\text{gen}} - t_k^{\text{sim}}|$ in seconds. Full-duration Wan and MotionCtrl tracks use their timestamp-mapped evaluation timeline.

Response Acc. For each simulator-responsive target, a 2D linear-velocity model is fitted to its robust center from 1.0 to 0.125 s before contact. Response magnitude is the median normalized residual from that extrapolation between 1 and 2 s after contact. A reference target is applicable only if its residual is at least 0.006. The generated response is correct when its residual is at least $\max(0.006, 0.5r_{\text{sim}})$. An observable target with no generated approach is a failure; insufficient visibility is excluded. Response Acc. is the fraction of correct applicable targets, expressed as a percentage.

Overlap Exc. For two tracked point sets, let

$$o_k = \frac{\text{area}(\text{hull}(P_k) \cap \text{hull}(Q_k))}{\min\{\text{area}(\text{hull}(P_k)), \text{area}(\text{hull}(Q_k))\}}. \quad (\text{B11})$$

The metric is $\max\{0, Q_{0.9}(o^{\text{gen}}) - Q_{0.9}(o^{\text{sim}})\}$ over the 2 s following each sequence’s visual contact. It is a dimensionless excess 2D hull-overlap fraction; zero is not evidence of zero 3D penetration.

Shape Err. and Area Drift. For each deformable object and frame with at least six jointly tracked points, the simulator and generated point configurations are independently centered and divided by their root-mean-square radius. The generated configuration is then aligned to the simulator configuration by the optimal 2D Procrustes rotation. Shape Err. is the mean corresponding-point distance after this normalization and is dimensionless. An object record requires at least eight valid frames and averages their errors.

On the same jointly visible points, let $a_k = A_k^{\text{gen}}/A_k^{\text{sim}}$, where each A_k is the tracked-point convex-hull area. With a_0 defined as the median of the first $\min(10, N)$ valid ratios, Area Drift is

$$\frac{1}{N} \sum_{k=1}^N |a_k/a_0 - 1|. \quad (\text{B12})$$

It is a dimensionless projected-area proxy, not a mass-conservation score, and also requires at least eight valid frames.

Recovery Acc. For each declared deformation interval, shape distance is measured relative to the last valid pre-event frame using the same Procrustes normalization. Let p be its 90th percentile during the event and r its 20th percentile from 0.5 to 2.5 s after the event. The recovery fraction is $\rho = \text{clip}((p - r)/p, -1, 1)$. An event is applicable when $p_{\text{sim}} \geq 0.015$, $\rho_{\text{sim}} > 0.05$, and both videos have valid measurements. It is correct when

$$p_{\text{gen}} \geq 0.5p_{\text{sim}}, \quad \rho_{\text{gen}} \geq \max\{0.10, \rho_{\text{sim}} - 0.30\}. \quad (\text{B13})$$

Recovery Acc. is the event accuracy, averaged event-first within sequence and then across sequences.

VideoPhy-2 AutoEval is separate from these paired metrics. SA expands to Semantic Adherence and PC to Physical Commonsense; both are custom-data VideoPhy-2 means on a 1–5 scale using the frozen shared prompts. Avg is the arithmetic mean $(SA + PC)/2$, not an official VideoPhy-2 aggregate. These scores are auxiliary perceptual diagnostics and do not verify the declared trajectory.

Effective support. Table 6 lists the method-specific supports for the deterministic metrics and the auxiliary VideoPhy-2 scores. For the first six metric columns, each cell is ordered as records/sequences/identities; the final three columns are ordered as sequences/identities. The counts are taken from the existing records, and identities are distinct input-image SHA-256 values.

Table 6 Effective support for simulator-trajectory adherence and auxiliary VideoPhy-2 diagnostics. “–” denotes an incompatible protocol.

Method	Collision Proxies			Deformation Proxies			VideoPhy-2 AutoEval (Auxiliary)		
	Contact Err.	Response Acc.	Overlap Exc.	Shape Err.	Area Drift	Recovery Acc.	SA	PC	Avg
LingBot-Cam	14/14/6	1/1/1	2/2/2	19/17/8	19/17/8	6/2/1	108/61	108/61	108/61
LingBot-Act	14/14/6	2/2/2	1/1/1	27/24/12	27/24/12	3/3/2	108/61	108/61	108/61
Wan2.1-I2V	14/14/6	3/3/2	6/6/5	31/26/14	31/26/14	11/5/2	108/61	108/61	108/61
Sora-2	14/14/6	7/7/3	8/8/4	31/26/14	31/26/14	4/3/2	108/61	108/61	108/61
MotionCtrl	14/14/6	8/8/3	4/4/3	31/26/14	31/26/14	–	108/61	108/61	108/61
Tora-Initiator	14/14/5	3/3/1	4/4/2	29/24/13	29/24/13	1/1/1	97/53	97/53	97/53
minWM-Wan	14/14/6	8/8/3	9/9/4	31/26/14	31/26/14	12/9/3	108/61	108/61	108/61
TourPhysics	14/14/6	5/5/3	9/9/5	30/25/13	30/25/13	8/7/2	108/61	108/61	108/61

The Contact, Response, Overlap, Shape, Area, and Recovery columns list records/sequences/identities. The SA, PC, and Avg columns list sequences/identities. Counts are after method-specific support filtering; “–” denotes an incompatible protocol.

B.4 Appearance and Long-Horizon Consistency

We evaluate input-scene subject consistency, background consistency, temporal stability, and camera-motion classification accuracy. TourPhysics obtains input-video subject consistency of 0.9743, input-video background consistency of 0.9791, and camera-motion classification accuracy of 0.8921 under the reported protocol.

Cache drift. For a decoded RGB video $I_0, \dots, I_{T-1}$, define the mean-channel luminance proxy

$$L_t = \frac{1}{3HW} \sum_{p,c} I_t(p, c). \quad (\text{B14})$$

It is the arithmetic mean of decoded 8-bit channels, not a gamma-corrected photometric luminance. Let $m = \max\{1, T - \lfloor 0.8T \rfloor\}$. The two drift measures in the main-paper memory-diagnostics table are

$$D_{\text{end}} = |L_{T-1} - L_0|, \quad (\text{B15})$$

$$D_{\text{tail}} = \left| \frac{1}{m} \sum_{t=T-m}^{T-1} L_t - \frac{1}{m} \sum_{t=0}^{m-1} L_t \right|. \quad (\text{B16})$$

Both are unnormalized 0–255 intensity-level differences; they are not percentages. The reported entries are equal-weight sequence means over the same 72 paired ordinary-cache and persistent-sink videos. These 72 sequences come from 27 distinct input-image identities. The memory router has nonempty causal revisit targets in 70 of those sequences, covering 26 identities, but all 72 sequences contribute to the cache-only drift comparison.

Historical-region error. The frozen appearance plan associates target latent j with an earlier source latent $h(j)$; their decoded frame indices are $4j$ and $4h(j)$. Both frames are resized to the memory-token lattice, and the earlier frame is remapped by the plan’s source-coordinate field. Let $M_{j,p}$ indicate a valid, visible, static, and depth-consistent historical correspondence and let $\kappa_{j,p} \in [0, 1]$ be its frozen confidence. With

$W_j(\cdot)$ denoting that remap, the base historical error is the confidence-weighted RGB Charbonnier error

$$E_{\text{hist}} = \frac{1}{\sum_{j,p} M_{j,p} \kappa_{j,p}} \sum_{j,p} M_{j,p} \kappa_{j,p} \cdot \frac{1}{3} \sum_c \sqrt{\left(\frac{I_{4j}(p, c) - W_j(I_{4h(j)})(p, c)}{255} \right)^2 + 10^{-6}}. \quad (\text{B17})$$

Thus, its region is exactly the routed historical-correspondence mask, and its base error is dimensionless.

Known-region error. Let $K_{t,p}$ be the simulator-consistent known-background mask (`reference_bg_lv`, with the frozen preserve/calibration-valid mask as fallback). It excludes the tracked dynamic foreground. Against the paired simulator RGB I^{sim} , the base error is

$$E_{\text{known}} = \frac{\sum_{t,p,c} K_{t,p} |I_t^{\text{gen}}(p, c) - I_t^{\text{sim}}(p, c)|}{3 \sum_{t,p} K_{t,p}}, \quad (\text{B18})$$

an MAE in 0–255 intensity levels, averaged over the memory-intervention chunks. This quantity is a guardrail diagnostic, as specified in the “Window-level quality gate” paragraph of the main-paper “Quality-Gated Atomic Commit” subsection; it is not an acceptance threshold.

The appearance-memory treatment and control use the same physical and camera trajectories, generator-facing depth, diffusion configuration, prompts, checkpoints, and random seeds. For a scene s , a paired percentage is $\Delta_s(E) = 100(E_s^{\text{mem}} - E_s^{\text{off}})/E_s^{\text{off}}$, so negative is favorable. History change is the equal-weight mean of $\Delta_s(E_{\text{hist}})$ in both subsets. The frozen development report also uses the mean of four paired $\Delta_s(E_{\text{known}})$ values. The held-out known-region guardrail uses the relative change of identity-macro MAE, $100(\bar{E}_{\text{known}}^{\text{mem}} - \bar{E}_{\text{known}}^{\text{off}})/\bar{E}_{\text{known}}^{\text{off}}$, which yields the reported -0.90% . Stating these frozen aggregations explicitly avoids conflating a mean of percentages with a percentage of means.

Memory subsets. The four-sequence development stress set uses four distinct input identities and covers rigid, cloth, and elastic content. Each selected sequence has six windows (481 output frames), 51 planned target latents, and active revisit chunks 3–5. Selection favored interpretable revisit/occlusion pressure, geometric-correspondence support, material diversity, and qualitative readability; earlier mechanism-screening metadata was available, so this is not an unbiased generalization set.

The held-out set contains all 14 eligible dynamic input identities unused by the mechanism pilots or development set. Camera-only identities are excluded, and one scene per remaining identity is chosen as the lexicographically first eligible six-window sequence satisfying the same 481-frame, 51-target-latent, chunks-3–5 contract. This identity-disjoint selection was frozen before treatment outcomes were inspected. Consequently, the main-paper memory-diagnostics table reports four development identities and 14 held-out identities, not repeated variants of a smaller image set.

C Additional Ablations and Limitations

C.1 Cache and Memory Ablations

The persistent-sink cache reduces long-horizon brightness drift compared with an ordinary finite temporal cache. On the long-video subset, end-to-start drift decreases from 26.33 to 11.17, while tail-to-head drift decreases from 20.38 to 7.74.

With the observation backbone and generator-facing depth fixed, geometry-routed appearance memory reduces historical-region error by 8.25% on the development split and 2.39% on the held-out split. Known-region error decreases by 2.99% and 0.90%, respectively. The smaller held-out improvement reflects variation in camera coverage and geometric correspondence quality.

The fallback property is verified by disabling all valid historical correspondences. In this case, $g_s(p) = 0$ for all p , and Eq. (A28) recovers the native observation path exactly. This test confirms that unavailable or unreliable memory does not introduce an additional visual bias.

C.2 Limitations

TourPhysics constructs a controllable physical hypothesis rather than a unique real-scene reconstruction. Its physical behavior depends on the supplied object proxies, material families, boundary conditions, and simulator parameters. Therefore, adherence to the prescribed simulation trajectory does not guarantee real-world physical accuracy.

The current geometry is incomplete in occluded or newly exposed regions. Appearance memory also excludes moving and deforming objects, which limits its ability to preserve object-specific appearance after large physical changes. Incorrect camera or depth estimates can further reduce valid cross-view correspondences.

Finally, finite-window diffusion is not real-time, and uncompressed historical pages grow with the number of accepted views. Future work should investigate fixed-capacity page eviction, object-local memory for deforming surfaces, real-world physical evaluation, and longer interactive tours.